NATURAL ENVIRONMENTS OF

THE STATE BOTANICAL GARDEN OF GEORGIA

By

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The purpose of this digital edition of Wharton's Natural Environments of the State Botanical Garden of Georgia is to give the document wider distribution.

I omitted the appendix listing the plants of the SBG natural areas. Such a list is already electronically available and is more current than Wharton's version.

I further reduced the size of Wharton's original report by re-formatting it. I scanned the original and changed the font size and spacing so that it would require fewer pages when printed. To improve readability I grouped the figures and moved them to the end of the document, just before the maps. I also corrected some typos. I formatted and re-captioned the tables to take advantage of Word 2007 features. I made content changes in the appendices on insects, amphibians and reptiles of the SBG. In the insect appendix I added or changed the higher level group names to reflect current taxonomic practice. For the amphibians and reptiles I provided the current scientific names for the taxa listed by Wharton and added species that are known to occur in the SBG, but were not mentioned by Wharton. Those are the only content changes I made.

I hope this document is made available to a large number of interested naturalists, both amateur and professional.

Dale Hoyt

Contents

ACKNOWLEDGEMENTS

Jeff Lewis and Jim Affolter, along with Anne Shenk, have made this study possible. Somehow they had seen a copy of my prior work at Elachee (Wharton 1994), and asked me to do a similar study of the 300 plus acres of The State Botanical Garden of Georgia (hereafter SBG). Jim Affolter was liaison and bore the brunt of my demands.

Many contributed their time and expertise. Giles Allard was most helpful in geology as were Larry Morris (School of Forest Resources), Steve Lawrence (NRCS), William Johnson (Cooperative Extension Service Soils Lab), and Owen Plank with soils. Dealing with plants was made easier by the help of Jennifer Ceska. The assistance of Mike Moore, and especially Jennifer Cruse at the University of Georgia Herbarium, was invaluable. Besides doing the bulk of identifying plants collected, Miss Cruse went along on several field trips. Elaine Nash and Anne Matthews contributed information on grasses and general botany. I am grateful to Bruce Bongarten of the School of Forest Resources, who took time to go through the manuscript and later to accompany me on a final winter field trip through the SBG.

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C.W.

MATERIALS AND METHODS

Soils were sampled with a footing spade at 38 stations at depths of 6" and 12" (data listed in Appendix 1). Sometimes a third subsoil sample was taken at 18- 24". Samples were routinely analyzed at the Soil Testing and Plant Analysis Lab (Ag. Services Lab). The data are presented in Appendix 1. Pounds/acre were converted to parts per million (ppm) by dividing by two. Millequivalents per 100 grams (meq/100 g) were calculated for calcium by dividing pounds/acre by 400 and for magnesium by 240. Cation exchange capacity (CEC) was calculated but did not seem to vary enough to be significant and was weakly correlated with the amount of bases in the soil.

According to the Natural Resources Conservation Survey (NRCS) of Clarke County, the bulk of the SBG soils are Madison sandy loam (15 to 25% slopes) and eroded. Most of the 6" samples were taken in the B 1 t zone, those at 12" in the B2t zone. Presumably our deeper samples would fall in the B3t zone if I interpret their survey correctly. The rest of the SBG is considered either severely eroded Pacolet sandy clay loam or Louisberg loamy sand (on the headwaters of South Creek). The two beaver ponds and most of the lower, wetter floodplain lie in Chewacla soils (CoB), while the slightly higher floodplain areas lie in Congaree soil (CoA).

If plants could not be identified with certainty in the field, specimens were pressed and identified by the UGA Herbarium. Animals were checked with the collections at the UGA Museum of Natural History.

To sample fauna a series of large plastic cups containing a water-formalin mix were sunk in the floodplain (N = 14), the slope west of the Purple Trail (10), the streamside zone and colluvial flats of South Creek (9), up ravines and in colluvial flats along Amphibolite Branch (9), North Bluff (5), coves and slopes between North Bluff and North Creek (20), and up the ravines of North Creek (21). The "pit- fall" traps were run weekly, then every other week and, in cold weather, at greater intervals. They were removed January 20,1997. On June 10,15 plywood squares ("critter boards") measuring 24" by 24" by ~" were placed along the ecotone of the Ivywood beaver pond and 17 were set along the ecotone between upland deciduous forest and floodplain. A synoptic collection of representative invertebrates and vertebrates has been prepared.

GEOLOGY

The SBG lies within a geologic zone called the Inner Piedmont. Allard and Whitney (1994) and Hall (1996) discuss in detail areas lying southeast of the SBG. Hall's study maps the southern end of Clarke County terminating about three miles northwest of the Barnett Shoals spillway. About 95 kilometers wide, the Inner Piedmont trends northeast, bordered on the northwest by the Brevard Zone (characterized by much faulting) and the Chauga Zone, characterized by phyllites, impure marble and augen gneiss which underlie the Elachee Nature Science Center (Chicopee Woods) at Gainesville, Georgia (Wharton 1994). The Inner Piedmont is bordered on the southeast by the Middleton-Lowndesville Fault separating it from the Carolina Terrane. I infer from the above studies that the SBG lies near the northern edge of the "core" area of the Inner Piedmont, the dominant rock types being either a Paleozoic-aged biotite gneiss or a migmatized biotite gneiss of Proterozoic/Paleozoic age with, in places, much lesser amounts of amphibolite, schist and quartzite.

The basic country rock of the SBG is a light-colored, grainy gneiss termed a paragneiss. Originally, perhaps 550 to 700 million years ago, this gneiss was thought to have formed of volcanic sediments (such as pyroclastic tuffs) later metamorphosed and folded in the Appalachian orogeny (episode of mountain building which also affected the Piedmont) during early and late Paleozoic times (Fig. 1). 7

The dominant rock of the SBG, called a migmatitic paragneiss or migmatitic biotite gneiss with associated schists (G. O. Allard, pers. com.) can be easily observed in the stream bed of South Creek along the Orange Trail (G1, Map 1). It is called a migmatite because due to an increase in pressure and temperature, component minerals with lower melting points (such as quartz and feldspar) became partially melted (neosome) within the original (relict) metamorphic rock (paleosome). The quartz/feldspar neosome often recrystallized as visible veinlets and nodules within the more or less modified paragneiss. If you closely examine the outcrops along the Orange Trail you can see the thin dikes, veinlets and nodules of light color interfingering with the more or less banded gneissic rock (Allard 1995). In places (G6, Map 1) the gneiss appears fine-grained and homogenous, being only slightly migmatized.

Along the Orange Trail and in the creek bed you can also find a very dark, almost black rock, called amphibolite. It was originally derived from basaltic lava and tuff or other mafic (rich in ferromagnesian minerals) rocks. Most of the black rocks you see in the creek bed will, however, not be amphibolite. Most will be gneiss with a surface coating of manganese oxide (G4, Map 1). Fig. 1 attempts to depict the origin and relationships of the dominant rocks of the SBG area.

Interest in the rocks of the SBG is more than academic. It so happens that since mineral soil is derived from the bedrock below, and plants can use whatever minerals are released from it by weathering, that the plant community may be thereby influenced by the rock type. their subterranean location.

SBG soils weather to a dull or deep red and appear darker than the C- horizon ("red clay") of typical Piedmont soils. This study seeks to relate the plant communities to the nutrient content of the SBG soils, particularly where agriculture has been limited or absent, such as on steep slopes adjacent to creeks or the Oconee floodplain. My study of the SBG soils (largely Madison) indicates that they are considerably more fertile than typical Madison and Cecil soils elsewhere in the Piedmont, in spite of previous agriculture and erosion. Since mineral soils are derived from the disintegration of bedrock via a residual saprolite (soft, decayed rock Showing original structure) their analyses can help us understand where the soil nutrients may have originated.

Biotite paragneiss analyzed by Woolsey (1973) contained quartz (42%), plagioclase feldspar (36-43%) and biotite mica (13%). Plagioclase feldspars in particular, can be a source of calcium and other minerals as they decompose. In addition, if the outcrops along the Orange Trail and up the tributary we call "Amphibolite Branch" are any indication, amphibolite layers, beds and lenses in the paragneiss may release considerable additional nutrients. According to Woolsey, Clarke County amphibolite contains 12.5% CaO and 8.5% MgO. If present, calc- silicate type quartzites could also release considerable calcium.

A piece of amphibolite from the Orange Trail (this study) was powdered and analyzed. It contained three times the calcium, 22 times the phosphorus, 33 times the zinc, and 6 times the manganese of the paragneiss sample analyzed by the same lab. Magnesium and potassium levels were similar. Powdered amphibolite (pH 7.0) and gneiss (pH 6.6) lie in the neutral pH range used by NRCS.

Amphibolite beds, as with the paragneiss, often bend sinuously, striking between N 100 E and N 500 E, making it difficult to know where they underlie the plant communities. Float (loose surface rocks) rock, however, often gives away

SOILS

Soil. pH and the plant communities

When we began trying to relate the SBG plant communities to the soils, I had hoped to find that the pH of the soil would be a reliable guide, since it is relatively easy to determine it by portable instruments. It was decided however to rely on the laboratory analysis, since instruments currently available have not been proven. Table 1 lists the NRCS pH categories, ranging from strongly acid to neutral, as determined by the laboratory. From this study it appears that pH is not as reliable a guide to the base content of a soil as is the content of calcium and magnesium. Lower calcium' levels may sometimes occur with higher pH. In the Elachee study (Wharton 1994) the highest level of calcium (3020) barely made it into the neutral category (pH 6.5), while a soil with less calcium (2545) gave a pH of 7.0. Nevertheless, portable instruments give a rough idea of the amount of basic ions in the soil.

The following are average figures for the SBG soils, beginning with the most acid, the heath bluff (pH 5.1), then dry oak-hickory on ridges (5.2), mesic oak- hickory (5.4), mesic oak hickory with chalk maple (5.5), beech-Ostrya (5.8), redbud (5.9) and redbud-mulberry (6.0). Note again that the common gneiss is 6.6 while the amphibolite is 7.0 (Table 1). Two tests with a hand meter (Kelway Soil Tester, Kel Instruments Company) were considerably higher than the lab results: 6.6 for a mesic oak-hickory site and 6.9 for a mesic oak-hickory with chalk maple.

Table 1. Calcium content of SBG soils.

Twenty seven soil and three rock samples grouped into four categories based on calcium content. Associated plant communities and NRCS categories are included.

Sample No.	Calcium (6" depth)	pH	NRCS category	Plant community
			CA < 100	
	97.4	5.5	strongly acid	beech-Ostrya
3	75.0	5.1	strongly acid	heath bluff
5	35.0	5.0	very strongly acid	dry oak-hickory
7	67.0	4.9	very strongly acid	heath bluff
9	98.3	5.1	strongly acid	heath bluff
13	94.8	5.1	strongly acid	dry oak-hickory
19	78.4	5.2	strongly acid	mesic oak-hickory
55	73.3	5.2	strongly acid	mesic oak-hickory
59	81.0	5.3	strongly acid	dry oak-hickory

Sample No.	Calcium (6" depth)	pH	NRCS category	Plant community
63	60.1	5.2	strongly acid	dry oak-hickory
73	98.4	5.1	strongly acid	mesic oak-hickory
			CA 100 -- 500	
11	336.9	5.5	strongly acid	mesic oak-hickory
17	376.1	5.3	strongly acid	bog
22	159.7	5.0	very strongly acid	chalk maple
33	194.1	6.0	medium acid	beech-Ostrya
35	277.8	5.6	medium acid	chalk maple
37	372.7	5.4	strongly acid	chalk maple
39	109.3	5.3	strongly acid	chalk maple
47	321.8	5.4	strongly acid	chalk maple
49	295.9	5.7	medium acid	mesic oak-hickory
51	206.2	5.3	strongly acid	mesic oak-hickory
53	155.3	5.6	medium acid	mesic oak-hickory
57	181.2	5.2	strongly acid	chalk maple
61	324.2	6.0	medium acid	chalk maple
65	328.0	5.7	medium acid	chalk maple
67	304.7	5.4	strongly acid	chalk maple
69	243.8	5.2	strongly acid	mesic oak-hickory
75	190.1	5.3	strongly acid	mesic oak-hickory
77	110.8	5.1	strongly acid	mesic oak-hickory
2R	231.8	6.6	neutral	gneiss rock (light color)
2R	224.4	6.6	neutral	gneiss rock, (dark, resembles amphibolite)
			CA 500-1000	
15	588.6	5.6	medium acid	mesic oak hickory
25	808.9	5.9	medium acid	redbud "patch"
31	766.7	6.1	medium acid	redbud-mulberry
				redbud-chalk maple
43	697.1	6.4	medium acid	mesic oak hickory
45	948.1	6.2	medium acid	redbud-mulberry

Sample No.	Calcium (6" depth)	pH	NRCS category	Plant community
				redbud-chalk maple
63A	637.0	5.7	medium acid	redbud-mulberry
				redbud-chalk maple
1R	766.9	7.0	neutral	amphibolite rock
			CA > 1000	
28	1689	6.0	medium acid	redbud-mulberry
41	1611.0	5.7	medium acid	chalk maple

Source of Soil Nutrients

In two out of three samples from the heath bluff the top 6" of soil is lower in calcium than the 12" depth. Both litter and bases would have trouble accumulating in surface soils of the steep, dry, rocky bluffs. I would guess that the calcium is coming from the weathering of bedrock here. In the other communities, 28 out of 36 samples had higher calcium in the top 6" (Appendix 1). It would appear here that calcium is derived from the litter, possibly augmented by fallout. In two out of four samples (samples 19 and 28) taken at a third depth (18-24") there was a further reduction in calcium with depth but, in two samples (redbud patch and chalk maple communities, samples 22, 25) calcium and magnesium increased below 12", suggesting a source from weathering bedrock.

In magnesium we see an opposite trend from calcium (Appendix 1). There is more magnesium in the deeper (12") layers (30 out of 36 samples). This suggests that magnesium is being derived from the bedrock below or is being concentrated by leaching of the layers above. If <u>both</u> calcium and magnesium increased with depth it would indicate that percolation and leaching were the probable cause, but this is not the case. This is a good place to point out that any soil possessing more calcium and magnesium than the rocks (Fig. 2) from which it was derived is experiencing bioaccumulation via the flora and fauna coupled with atmospheric fallout.

Relating plant communities to the content of the bases calcium and magnesium

Table 1 and Fig. 2 will help to summarize the relationship of plant communities to the two bases, calcium and magnesium. Mesic oak-hickory communities with a chalk maple understory (called "chalk maple" for short) appear, with one exception, to have calcium between 100-500. Communities heavy with redbud as a dominant understory have higher calcium levels (mostly between 500- 1000). The highest calcium recorded (1689) was also in this community. The presence of mulberry or chalk maple <u>together</u> with redbud is almost certainly an indicator of <u>comparatively</u> high calcium in the top 6" of the soil. One redbud "patch" also had a high (809) calcium. The presence of an <u>occasional</u> redbud apparently does not indicate higher calcium.

Botanists have long sought to correlate plant communities with some measurable soil parameter. This study tried to find evidence pro or con. Table 1 indicates that, if you consider only calcium (at 6") both the heath bluff and the drier hilltops (dry oak-hickory) are characterized by very low «100) calcium levels. Communities with a strong chalk maple component in the understory are definitely higher (100-400) in calcium. Under a mesic oak-hickory canopy, a combination of redbud-mulberry, redbud-chalk maple, or a redbud "patch" in the understory indicates much higher (600-1000) calcium levels. Mesic oak-hickory lacking these understory dominants spans a wide range of calcium (100-600). It is therefore difficult to separate mesic oak-hickory communities with and without chalk maple understories using either calcium or pH in the top 6".

However, if we average the calcium from both the 6" and 12". depths, a different picture emerges. Chalk maple sites (Map 2) are considerable richer than the sites of mesic forest without chalk maple (630 versus 424). Likewise, if we average the magnesium from both levels the same difference appears (412 versus 276). Calcium and magnesium levels are both 33% higher where chalk maple forms a dominant understory.

In conclusion it can be stated with some assurance that plant communities at SBG do indeed reflect the amount of calcium and magnesium in the soils. (So few samples are from' Beech-Ostrya and bog sites that comments are unwarranted). If small subcanopy trees such as chalk maple, redbud and mulberry are getting their mineral bases from deeper in the soil profile, then sampling only the top 6" could be misleading. The top 6" would probably affect tiny tree seedlings and certainly the herbaceous community. Three samples, the deepest of which reaches into the subsoil, would be ideal to verify the mineral source.

Levels of calcium and magnesium in SBG soils compared with other Piedmont soils

Perkins (1987) lists forested (either in hardwoods or hardwood pine) soils from the Georgia Piedmont. Table 2 compares the calcium and magnesium content ranked by calcium content. The four "poorest" soils are Madison and Cecil over granite gneiss and mica schist. The SBG Madison soils are more "fertile" with considerably more of both calcium and magnesium, having 120 times more calcium than the poorest forested Madison reported by Perkins. Yet the SBG soils have only 29% of the calcium of the remarkable Elachee soils, which in turn, are quite close in values to the dark base-rich rocks high in ferromagnesian minerals which underlie soils such as Iredell derived from diorite or gabbro or a residuum of mixed gneiss, amphibole and gabbro. In spite of their eroded nature, nutrient levels in the SBG soils are quite high. This fertility must lie in the nature of the paragneiss or the associated schists, amphiboles and perhaps even calcium-rich quartzites (calc-silicates). For comparison, a high magnesium limestone soil (knox dolomite) is included, supporting oak-hickory at an ISP site at Oak Ridge, Tennessee (Cole and Rapp 1981).

Table 2 Comparison of calcium and magnesium content among soil types.

Averaged content of calcium and magnesium levels in SBG soils (Madison, 37 samples) compared with levels in soils at Elachee's Chicopee Forest (18 samples) and with other forested (chiefly deciduous) Piedmont soils (Typic Hapludults) and Davidson soils (Rhodic Paleudults) from Perkins, 1987. Figures are in milliequivalents per 100 grams.

Soil Type	Parent rock	Ca	Mg	No. samples
Madison	over granite gneiss/mica schist	.05	.25	1
Cecil-	granite and granite gneiss residuum	.10	.04	1
Madison	mica schist residuum,	.18	.17	4
Cecil	granite and granite gneiss residuum	.28	.12	4
Madison	(SBG - present study) - paragneiss and schist with layers of amphibolite and quartzite	.66	.69	37
Pacolet-Tallapoosa series (15-45% slopes) (Elachee Nature Science Center)	mica/graphite/hornblende schists, gneiss, metagreywacke and phyllite	2.29	.80	18
Davidson	basic rocks high in ferromagnesian minerals (diorite, gabbro, amphibole, etc.)	2.40	.85	3
Typic Paleudult	Knox dolomite (IBP site, Walker Branch, TN	7.90	39.90	1

There is considerable difference between the flora of the SBG and the Elachee Nature Science Center (ENSC). There is a higher plant diversity at Elachee, some species being quite rare and some of montane affinities. Its rocks are more diverse and suggest an origin as sediments in an ancient marine delta, since metamorphosed and highly contorted (Wharton 1994). Elachee lies in the Gainesville Ridges, a physiographic subtype within the inner Piedmont. Elachee soils are essentially uneroded and have long, steep slopes. Most of the slopes are medium acid with calcium levels of 500-1000 supporting rich herbaceous communities, while truly lush herb growth is found on neutral sites with calcium levels from 2000-3000. Redbud frequently occurs in the understory. Part of the Elachee richness may be in part due to the increased moisture in its soils, particularly on the toe slopes and in ravines.

CLASSIFICATION OF ENVIRONMENTS

The wetland communities are quite different in species composition from the uplands and from one another. Differences between upland environments are often subtle and overlapping. The upland area of the SBG seems to be a mesic forest with little pronounced variation in canopy tree species. Species like pallid hickory and southern red oak suggest a drier phase of the typical ridge and slope forest. There seem to be no areas dominated by truly xeric species such as blackjack oak. In spite of their abuse by man the SBG soils appear to be <u>comparatively</u> well endowed with basic nutrients compared to other common Piedmont soils (see section on soils). The tentative classification which follows is a mix of botanical, physiographic and hydrological factors. Appendix 2 lists sites where significant communities, plants or other features were found. Most of these stations are indicated in Map 3.

The dominant Piedmont forest type is "oak-hickory," generally considered a deciduous forest. Although there are usually scattered shortleaf pines in the canopy, they are not common enough to warrant a deciduous-evergreen category; seedlings may occupy blowdowns and canopy gaps, particularly on the drier sites. The SBG is somewhat unique in the abundance of hickories. Perhaps selective logging of the oaks, ash and other species has favored their proliferation. Some of the oaks and hickories are troublesome to identify in the field. Try to look at the upper "sun" leaves and not the lower "shade" leaves. Scarlet oaks have narrow "aluminum" striping, dead lower limbs, a characteristic ringed acorn and smallish, deeply dissected leaves. Northern red oaks have wide "aluminum" striping and their bark appears much darker, their acorns are different and they tend to avoid the ridgetops where scarlet and southern red oaks predominate. Black oak and northern red leaves can be confused but the bark should help distinguish them.

The leaf shapes of white oak, post oak and southern red oak are distinctive. Water oaks on the SBG appear to be confined to the uplands, particularly in disturbed areas such as the former parking area just north of the gatehouse. About 100 feet or so down the <u>old</u> trail (now closed) and across the ravine to the north are two huge oaks, one a typical water oak, the other a white oak. (Vicinity Sta. 76, Map 3).

Even most botanists dread hickories. The SBG has at least four kinds. The most common appears to be a hybrid between pignut *(Carya glabra)* and sweet pignut *(C. ovalis).* This widespread tree Dr. Bongarten calls a "hybrid swarm." Bark-wise the interwoven parabolas flatten out and tend to flake off whereas in the mockernut the bark remains tighter. The small nut and the thinner husk of the hybrid is distinctive (the nuts are relished by squirrels). The larger nut of the mockernut with its very thick husk is diagnostic, as well as its larger, scruffy, pungent leaflets. Just north and a few feet off the "old" trail from the closed parking lot north of the main gate (Vicinity Sta. 76, Map 3) is a fine example of the rare to occasional bitternut hickory. The <u>very thin</u> husk on the smallish nut is distinctive. The sulfur-colored buds are hard to see unless a twig has fallen. Since the trail guides often use common plant names, common names (although highly variable) are used in the following classification. Appendix 3 lists common names alphabetically, followed by the scientific name. Refer to the botanical name to avoid confusion. Appendix 3 also lists plants alphabetized by botanical name.

NOTE: Letters in parentheses precede each common name where known or relevant: (a) abundant, (c) common, (0) occasional, (r) rare, (vr) very rare, (d) dominant, and (v) vine.

Uplands

Mesic oak-hickory forest (dry phase)

Occupies drier ridge-tops. Uncommon. Example: where Green Trail crosses ridgetop road. An acid and relatively uncommon environment.

<u>Canopy:</u> (a) scarlet oak, (c) southern red oak, (0) black oak, (0) winged elm, (0) post oak, (0) shortleaf pine, (0) mockernut hickory.

<u>Subcanopy:</u> (c) persimmon, (c) sourwood, blackgum, (0) dogwood

<u>Shrubs:</u> French mulberry, hawthorn, sassafras, blue haw, sparkleberry, winged sumac, dwarf blueberry, deerberry, common indigo bush

Herbs: St. John's wort, pussytoes, spotted wintergreen, wild quinine, yellow jessamine, ebony spleenwort, muscadine, bracken fern, silver bluestem, Elliott's Indian grass

Mesic oak-hickory forest (mesic phase) Ridges, slopes, ravines, flats.

Heath bluffs (Map 4)

This environment occupies steep, riverside bluffs along the Oconee River, with considerable exposed bedrock and boulders. The evergreen heath, mountain laurel, is diagnostic. The soil is low in bases and acid. The canopy is a mix of mesic (beech, red maple) and dry-adapted species (sourwood, pale hickory, mountain laurel and several herbs). Because surface soils dry rapidly and there is little humus, many herbs reflect dry conditions. Some, such as galax and Michaux's lily do not seem to occur elsewhere (the lily does occasionally occur in other environments). The two bluffs, south and north, are listed separately.

South bluff

Canopy: (d) white oak, (c) pale hickory, (c) beech, (c) red maple, (0) N. red oak

Subcanopy: (0) persimmon, (0) sourwood, (0) witch-hazel

Shrubs: (a) mountain laurel, (c) Elliott's blueberry, (0) horse sugar, (0) Oconee azalea*, (c) sweet shrub, (r) dog hobble (*I.D. by Steve Bowling. Needs verification when it blooms. The other possible I.D. is piedmont azalea.)

Herbs: (c) hawkweed), (c) Georgia mint, (c) St. Andrews cross, (0) galax, (0) featherbells, (0) Michaux's lily, (0) bird's fodt violet, (0) yellow jessamine, (0) xeric elephant's foot, (r) false fox glbve, (c) coreopsis

North bluff

Canopy: (c) N. red oak, (c) red maple, (c) white oak

Subcanopy: (c) sourwood, (c) black gum, (c) witch hazel, (c) chalk maple

Shrubs: (0) horse sugar, (a) mountain laurel, (c) maple-leaf viburnum, (0) azalea sp., (0) strawberry bush

Herbs: Greenbrier (S. glauca), (0) flypoison, (0) featherbells, alum root, (0) Michaux's lily, lion's foot, devil's bit, cranefly orchid.

Ridges and slopes (mesic phase, oak-hickory)

This category covers the bulk of the SBG. Soils of these mesic slopes vary widely in content of calcium and magnesium. Soils are medium to strongly acid. This environment has been heavily logged and, in places, in agriculture. It is classified by the NRCS as eroded to severely eroded. In places gullies that formerly drained row-crop agriculture are obvious and old terraces can occasionally be seen.

<u>Canopy:</u> (d) white oak, (0) N. red oak, (0) black oak, (0) shortleaf pine, (0) shagbark hickory, (c) mockernut hickory, (a-c) pignut-sweet pignut hickory hybrid

<u>Subcanopy:</u> (c) hop hornbeam, (c) dogwood), (c) sourwood, (0) red maple, (0) blackgum, (0) black cherry, (0) American holly, (0) serviceberry

<u>Shrubs:</u> (0) dwarf hackberry, deerberry, Elliott's blueberry, cockspur hawthorn, dwarf paw paw, azalea sp., upland possum haw

<u>Herbs:</u> Canadian avens, muscadine, buttercup, wild rye, Virginia creeper, pale Indian plantain, snakeroot (two species), honewort, wild ginger, pokeberry (in patches), rattlesnake fern, black-seeded needle grass

Ravines and lower (toe) slopes (Map 4)

These are rich, moist environments with a high species diversity. Here, also, we find the most basic, circumneutral soils (NRCS categories medium acid to neutral), and areas of subcanopy dominated by chalk maple (which, like beech, seems to retain its dried up leaves in winter) and other base-loving plants like red bud. Beech, mulberry and a host of shrubs and herbs thrive in this environment which receives both water, and possibly nutrients, from upslope (see Map 2). These environments share a high plant diversity with the colluvial flats.

<u>Canopy:</u> (c) beech, (c) N. red oak, (c) white oak, (0) bitternut hickory, (0) basswood, (c) red maple, (r) cucumber tree, (0) white ash, (0) winged elm

<u>Subcanopy:</u> (0) redbud, (c) chalk maple, (c) serviceberry, (c) silverbell, (c) mulberry, (0) bladdernut, (0) hop hornbeam, (r) hog plum, (r) umbrella tree, (r) possum haw

<u>Shrubs:</u> (c) dwarf paw paw, (0) sweet shrub, (c) painted buckeye, (r) Georgia hackberry, (r) buckthorn bumelia, (r) Carolina buckthorn, (r) black haw, (0) wild hydrangea, (0) Elliott's blueberry, (0) maple-leaf viburnum, strawberry bush

<u>Herbs:</u> (a) black snakeroot, sessile bellwort, wild onion, (c) Canadian avens, (0) agrimony, (0) Curtis' gOldenrod, devil's bit, (c) carrion flower (two species), alum root, (0) jumpseed, Jack-in-the-pulplt, (0) bloodroot, perfoliate bellwort, wild geranium, wild yam, thimbleweed, poison oak, Mack's honeysuckle, (0) coral honeysuckle, rattlesnake root, (c) Christmas fern, (0) maidenhair fern, (0) round-lobed hepatica, (0) lovage, Carolina phlox, windflower

Stream edge zone and colluvial flat

These are moist "almost wetlands." The Orange Trail proceeds along the stream edge zone, as does the lower side trail up "Amphibolite Creek" (at the bench). Small flat areas close to the creeks are formed by non-alluvial (not deposited by a permanent body of flowing water, as a floodplain is formed by a river) means, largely by soil creep, freeze and thaw phenomena and downslope sheet erosion. Some plant species are naturally shared with the previous environment. Several species of ferns and composites, yellowroot, eleagnus, and silky dogwood are more or less confined to this habitat.

Canopy: (c) tulip poplar, (c) white oak, (0) slippery elm (a potential canopy species but only small examples have been found)

Subcanopy: (c) ironwood, (c) silky dogwood, (c) mulberry

Shrubs: (c) azalea (several species), (c) painted buckeye, (0) eleagnus, (0) strawberry bush

Herbs: (c) yellowroot, (c) coneflower, (c) Canada snakeroot, (c) climbing hydrangea, (0) lion's foot, (0) cross vine, (c) lopseed, (c) common wild pea, carrion flower, (0) river cane, (0) mayapple, (0) wild yam, (0) bloodroot, Jack-in-the-pulpit, rue anemone, (0) jumpseed, (r) green dragon, (c) lady fern, (0) broad beech fern, (r) cinnamon fern, (0) netted chain fern (at bridge over' first branch north of beaver dam on Orange Trail), knotweed, angelica.

Successional forest

These grow on former fields and pastures. Remnants of old terraces and erosion gullies can frequently be seen. Topographic maps reveal these areas as having comparatively gentle slopes. Stands of pine (normally loblolly) are obvious. Late stages of succession are indicated by old loblolly pines here and there among the young to intermediate age hardwoods. Sometimes you can find the stumps where the old pines were cut out. (An occasional shortleaf pine was apparently normal for many Piedmont hardwood forests.) It normally takes about 100 years to restore the hardwood forest. The lost topsoil was formerly about a foot thick. If a mature hardwood forest could rebuild the organic-rich topsoil at the rate of one inch every 50 years it would take at least 600 years to rebuild the topsoil lost by row-crop farming. There are successional changes in animal life as well, from harvest mice who eat seeds to deer mice that eat acorns. The bird community also changes dramatically from things like certain sparrows, towhees and indigo buntings to forest dwelling species like thrushes, great crested flycatchers and Kentucky warblers.

Canopy: (c) loblolly pine, hardwood species

Subcanopy: (0) eastern red cedar, hardwood saplings and pole-stage trees

Shrubs: (c) sparkleberry, (c) blueberries, (0) shrubby serviceberry

Herbs: greenbriers, Japanese honeysuckle, ebony spleenwort, ground pine, bracken fern, plume grass

Wetlands

Floodplain hardwood forests

Two types of floodplain forest can be recognized. One is a "hydric phase," a very wet environment on

Chewacla soils (Map 5). This is the type directly below the Callaway Building. West of the trail crossing, it may be abnormally flooded by runoff from the once-forested area around the Conservatory, possibly ponding for periods during the crucial growing season. The area should be closely monitored for stress on the canopy species. A good example of this wetland forest lies east of the trail crossing, essentially an ash-elm-birch-box elder forest totally unlike any other environment.

The White Trail follows a natural levee along the Middle Oconee River. Native shrubs and subcanopy trees have been largely eliminated by Chinese privet (seeds carried by birds). The privet should be destroyed here.

The other type of floodplain is found between the maintenance sheds and the band of Chewacla soils lying just inside of the levee (and White Trail). Here is a band of comparabie width described as Congaree soil (Map 5), slightly higher and moderately well-drained. The eastern end of this strip of Congaree soil extends in an expanded bulge towards the powerline. This area is now in succession. It originally bore a canopy of moisture-loving trees, some of which are not typical of floodplains, such as yellow poplar. Species include red maple, sweetgum and perhaps elm and box elder. It is subject to inundation less frequently and for shorter periods of time than the hydric floodplain--a "meso-hydric" floodplain community. A similar community was described at Elachee and supported, in addition, hackberry, paw paw, black walnut and honey locust. The levee itself is a better-drained version of the hydric floodplain. Dominant are box elder, river birch and green ash. Actually box elder and river birch act as pioneer species on Piedmont floodplains, colonizing disturbed floodplain areas and the naturally-disturbed levees. There are some grasses (melic grass, river oats, lowland panic grass) owing to the barrenness and sunlight of the river bank. Green ash and American elm are classic bottomland hardwoods (climax forest species) on the lower terraces. On the highest floodplain terraces we could expect swamp chestnut oak, water oak, cherrybark oak and even loblolly pine. Vines are common. See Map 6 for areas of bottomland hardwood forest, wet ponding areas of hydric phase BLH, and areas of loblolly pine which followed aqriculture on the higher parts (Congaree soil).

Canopy: (d) green ash, (c) American elm, (c) river birch, (c) box elder, (0) sycamore

Subcanopy and shrubs: Other than a few ironwood and wetland possum haw, hydric floodplains have few native species. They are thus apparently vulnerable to invaders such as Chinese privet.

Herbs: (0) jewelweed, (c) common edible violet, (0) netted chain fern, (0) day flower, (c) cinnamon vine or air potato, (0) Small's greenbrier, (0) lizards tail (in very wet spots), (0) trachelospermum, (c) swamp nettle, (c) wingstem (tall composite), (c) crown beard (tall composite), (0) wild lettuce, (0) passion flower, (r) Carolina rose, (0) wood reed, (0) knotweed, (0) climbing hempweed, (0) mesic elephant's foot, (0) bluet, (0) honewort, (a) vine mint (in patches). On Levee: wild rye, river oats, melic grass, wetland panic grass, trifid beggar ticks.

Bog

Only one bog has been located (southeast of station 100). It is on a side branch that forks off near the base of the wooden "stairs" on the White Trail. There is a dense growth of river cane in this area. Since it is down below drainage from the toxic waste dump, there are several "wells" of PVC pipe in the vicinity for monitoring pollution. The bog is quite wet with a dense growth of sedges, rush and spikerush. It is a good place for salamanders (look under logs and bark, etc.).

Canopy: None

Subcanopy: Silky dogwood

Shrubs: None

Herbs: (a) climbing hydrangea, (c) lady fern, (c) bead fern, (0) spikerush, (c) sedge, (0) swamp nettle, rush, (c) common wild pea.

Beaver Pond

This wetland area (19.3 acres owned by the SBG) was acquired in 1990. In a 1994 study (Bolton et al), it was described as follows: scrub-shrub (16.2 a), emergent aquatic plants (2.4 a), and open water (1.2 a). It occupies Chewacla soils adjacent to McNutt Creek and the middle Oconee. It is accessed from a curve in Ivywood Drive at the property of Mrs. Westfall, who keeps part of the University property in grass, as well as maintaining a strip some 30 feet wide between her fence and the pond edge or ecotone. This makes it easy to observe the pond emergents and ecotonal vegetation at the pond edge. It is said that beavers dammed several small tributary creeks and created the pond in the mid-1940's. Many logs and stumps of the wetland forest still remain and make canoeing difficult. The pond level fluctuates as much as 6 feet, sometimes trapping fish behind Mrs. Westfall's fence.

Canopy: We assume none. Bolton et. al list at least three potential canopy species (red maple, river birch, green ash). They could be scrubby, dwarfed examples that qualify for the scrub-shrub category, an environment which I did not examine.

Subcanopy: tag alder, silky dogwood, black willow, box elder

Shrubs: elderberry, Hawthorne No.2, Chinese privet

Herbs: Emergent/submergent: (c) rush, (0) spikerush, bulrush, arrow arum, (a) burweed, duck potato, rush No.1, (c) cattail, (a) parrot feather Ecotone (pond edge): with occasional willow, silky dogwood, hawthorn, (a) (v) groundnut, beadfern, (0) (v) climbing hempweed, common edible violet, sedge No.1, swamp nettle, jewelweed, spikerush, (a) lizard tail, sedge No.2, (0) (v) virgin'S bower

WHY ARE PLANTS SUCH AS TRILLIUMS MISSING FROM THE SBG?

It might be rewarding to contrast the results of this study with that of a study of the Elachee Nature Science Center (ENSC) acreage on Walnut Creek, described by Wharton (1994). The plant life of the SBG is markedly different from that of the ENSC. Part of this difference may be explained by the fact that Athens lies in the "midland" or "inner" Piedmont while Elachee lies in the Gainesville Ridges, a narrow series of ridges running SW-NE from northern Stephens County to Atlanta and offering quasi-montane microclimates. The Chattahoochee runs along its western boundary and before the famous "stream capture" at Tallulah Falls, the Chattooga was its headwaters.

While Elachee is only 25 miles north of the SBG it is very close to this biotic corridor draining the entire northeast Georgia mountains and part of North Carolina. This may account for more species with montane affinities at Elachee than at the SBG. Some of these that are missing from the SBG are: silvery glade fern, goat's beard *(Aruncus)*, toothwort *(Dentaria)*, black cohosh *(Cimucifuga)*, horsebalm *(Collinsonia)*, yellow mandarin *(Disporum)*, cucumber root *(Mediola)*, ginseng, lousewort, New York fern and spikenard *(Aralia racemosa).* The absence of trilliums at the SBG is anomalous (there are four species at Elachee).

Other notable absences are those of spicebush *(Lindera)*, hazelnut, buffalo nut, and storax among the shrubs. Among the trees which I did not find at the SBG are mountain camellia, American chestnut, white pine and Virginia pine, nor have I found walnut or honey-locust on the floodplain. Bruce Bongarten states that spicebush, hazelnut, storax, American chestnut, black walnut and honey locust can be found naturally occuring within two miles of the SBG. Both he and I are at a loss to account for their absence at the SBG. I have seen abundant trilliums *(Trillium catesbei, T. cuneatum)* in deciduous forest in extreme southern Oconee County off the Colhan Ferry Road near the Greene County line and Van Giesen (1995) recorded both these species near the south side of Athens, so these two species at least, should occur at the SBG.

Can their absence be attributed to agriculture, grazing, excessive logging and erosion from the heavily used upland slopes? It seems more logical that other human activity, such as rooting hogs or possibly "collectors" or "bulb diggers" from the very heavy foot traffic through the area would be responsible. The absence of the valuable medicinals, ginseng and goldenseal might be expected.

It could be that, because the SBG is further south and has less precipitous terrain with less microniches, montane species have not been able to survive as they have at Elachee. Bluffs and ravines along the several forks of the Oconee should be investigated, such as the beautiful north-facing high bluff across the river from the southern boundary of the SBG. Possibly the North Oconee has more suitable microniches. I say this because Steve Bowling and I were amazed to find a stand of *Trautvetteria carolinensis* on a tiny creek in a bluff of the North Oconee (Gary Nelson property) just a mile or so southeast of the SBG. This is a plant of the high elevations, seldom seen below 3000'.

MAN'S IMPACT ON PIEDMONT FORESTS

Caucasian settlements built westward to the Oconee by 1784, and extended west to the Ocmulgee by 1805. Row crop farming and erosion that accompanied it, rather quickly depleted soil fertility and it was reported that as early as 1850 wagon loads of settlers left the "worn out" Piedmont soils for the "new" soils of Alabama and Mississippi. While at first subsistance farming of edible crops such as corn was the rule for the early settlers, abandoned small farms apparently became consolidated into plantations dedicated to cotton monoculture as the dominant land use.

Human and animal labor were the dominant energy inputs, although fertility was temporarily boosted by the importation of Chilean nitrate. Large plantations broke up with the loss of slave labor. Piedmont lands were "abandoned" in three phases: during and following the Civil War, during the agricultural depression ofthe 1880's and following the boll weevil of the 1920's. The secondary forest that followed was itself exploited three times: lightly until 1920, severely during World War II and steadily since the mid-1940's by pulpwood harvest (Wharton 1977).

Thus there was an era (1800-1930) when Piedmont soils, which probably took at least a millennium to form, were lost in a <u>short term</u> and unsustainable production of wealth using animal and human (slaves, sharecroppers) labor. But the <u>long term</u> productivity of Piedmont lands was gone, along with approximately 12 inches of topsoil. People had to gravitate to cities to survive.

> The sharecrop cabins are leaning awry
> Their clapboards weathered and grey
> And the chinaberry trees weep in the wind
> For the people who went away. *
>
> (* From "The Rust Red Earth" by the author)

Some of the old cotton lands were planted to pines. With the aid of machinery and fertilizer and lime to counteract the acid soils, pastures came to also occupy much of the terrain. Now we have the vista of artificial "prairies" (pastureland) to produce protein (cattle) from grass where nature intended that carbohydrates be produced (chiefly ceuulose, but including fruits, nuts and berries) by forest vegetation. It is, of course, costly to go against nature's plan and without the large input of fertilizers, minerals and oil-powered equipment it would not be possible.

Map 7 indicates the extent of man's modification of the SBG as near as could be construed from an aerial photograph made about 1938. Note that the Congaree soil of the meso-hydric floodplain was in agriculture at the time (compare with Map 5). Map 8 shows where, on the upland, pines and pine-hardwood occurs according to a map prepared by William Seery. The presence of pines could indicate other catastrophic influences, from over-zealous logging of hardwoods to fire, storm or livestock damage. Map 9 shows where pines are presently conspicuous in the forest canopy, along with the extent of open land, a "new," quasi-prairie environment.

Finally, Map 10 points out some troubled areas, several needing close monitoring such as part of the floodplain directly below the Callaway Building. The large privet thicket east of the trail crossing the floodplain should be eliminated by appropriate herbicides. Quadrats should be established in dense Japanese honeysuckle and these sprayed experimentally in the spring before other herbs emerge.

Map 1 indicates the presence of stone piles along the Orange Trail. Such piles occur throughout the mountains and at places in the Piedmont. They have even been observed in South America (Giles Allard, pers. com.). I have seen them near the Chattahoochee River (Habersham County), at Poole Mountain (Gwinnett County) and along Beech Creek and Eagle Fork Creek (Clay County, North Carolina). Most of these probably represent pre-cotton era subsistence agriculture.

Often the rocks are piled around the base of tree stumps or large unmovable boulders. I expect that, initially, entire families participated in this means of clearing stony ground. Most stone piles were eliminated by the construction of terraces for cotton monoculture. Here, on the Orange Trail, we can see them just below the terrace system. Stone piles are frequently thought to be "Indian graves," but without skeletal remains or artifacts from excavated piles, evidence points to the manual labor of clearing early fields for subsistence farming.

MANAGED VERSUS UNMANAGED FORESTS

It is educational to examine a "virgin" or old growth climax forest. Admittedly, there aren't many samples to look at. There are many down trees (logs) and the amount of sunlight striking the ground in openings between trees is astonishing, especially when contrasted with the dense, second growth forests following heavy logging or colonization after natural disturbance.

In original forests almost everywhere, two factors predominate. One is that each kind of tree has its own life expectancy--some trees live much longer than others. Some species may live 75 years, others 300. Secondly, the old growth forest is an uneven aged forest. When trees reach their life expectancy, or are blown over or hit by lightning, they create natural openings or "gaps" which may occupy as much as 10% of the forest. These openings provide ample successional stages of vegetation to support a diversity of both plants and animals such as birds. Agriculture or clear-cutting changes all this. Subsequent forests are "even aged" and are seldom allowed to reach their primeval splendor. They are more "productive" of cellulose but not necessarily acorns, and are harvested periodically. Minerals like calcium in the trunks are removed, instead of slowly returning to the soil to be recycled.

In pre-Caucasian times there was sufficient vegetation in all stages of succession to provide ample food and micro-environments for a diversity of animal life. In the "young" managed forests of today artificial gaps, called "wildlife openings," are made by machinery. Harvesting by clearcutting is often defended as necessary to provide early successional stages which the old growth forest provided naturally. Large clearcuts or too many small ones tend to unbalance the foodwebs of the forest and population eruptions occur. Too many deer can become a problem. Soils become overheated and the soil fauna reduces organic matter too rapidly and controls, such as salamanders, are reduced or eliminated by the "new" microclimate. Species diversity may temporarily increase and some species such as in the insectivore guild, including shrews, may flourish.

Management questions should deal with long term sustainability. How long does it take for reductions in litter and soil organic matter, and the removal of large amounts of calcium in the trunks of trees to affect the health and productivity of the forest? The comparatively short lifetimes of humans makes long range study difficult, but there are indications from Europe that three intensive harvests significantly reduces tree growth.

ANIMAL LIFE AT THE SBG

INVERTEBRATES

Insects

Beetles (Appendix 4), particularly the predatory carabid ground beetles, along with the camel crickets, are the most numerous insects which crawl through the litter of the forest floor. The scarabs, many of which are dung beetles, are also common, appearing from late July on through October. The most conspicuous is the large black tumblebug *(Deltochilum gibbosum).* There are sporadic "outbreaks" of other species. In mid-August on the north bluff, about 50 specimens of a tiny (3-4 mm) metallic green scarab *(Glaphyrocanthon viridis)* suddenly appeared. There were similar sudden appearances of other scarab beetles: small (7 mm) scarabs *(Onthophagus hecate)* in mid-August along the ravines of North Creek, a slightly larger (9 mm) black scarab *(Copris minutus)* in mid-September on the slopes west of the

Purple Trail and a beautiful green scarab *(Geotrupes splendidus)* in late September-early October along the North Creek ravines.

The most common beetles are the mostly predatory carabids, of which four species were taken. Orange-legged forms of medium size (16 mm) *(Sclaenius aestivus* and/or *Harpalis* sp.) occur in every environment and are common from May through October. They are most abundant on the floodplain. A large (25 mm) black carabid *(Dicaelus dilatatus)* is widespread, except in the floodplain. Two other medium-sized ground beetles, *Evarthrus sigillatus* and *Pterstichus* sp., occurred in the moister environments.

Next to the ground beetles, the most common ground-crawling insect in our traps were three species of camel crickets *(Ceuthophilus graciles* [?], *Ceuthophilus* No.2 and NO.3). They are all small in the spring. By mid-summer large (10 x 40 mm) individuals with antennae over 60 mm long are present. They begin to appear on the floodplain by mid-June. A second "crop" of young individual appears in mid-September. Camel crickets are common to abundant in all environments. These herbivore-detritivores must form a good food source for predators. Since you seldom see them in the daytime, you would hardly suspect their abundance. One other seemingly rare orthopteran is the shield-backed grasshopper *(Alanticus gibbosus).*

Other beetles occurred occasionally in our traps. One, the weird-looking elongate rove beetle *(Platydracus maculosus)* appeared only in June and two relatively common long-horned beetles with wood-boring larvae (Cerambycidae: *Prionus imbricornis* and *Othosoma brunneum)* appeared between mid-June and mid-August.

Aquatic Life

We observed a few minnows (unidentified) in both North and South Creek. They are not common. Pollution may be a factor. Pollution is not at critical levels for we found the larvae of mayfly, stonefly and caddis fly in both North and South Creek, but principally in the rocky section of South Creek along the Orange Trail below the fork with Amphibolite Branch (at the bench). Appendix 5 lists common taxa in South Creek or the "Orange Trail stream" as identified by Bob Hall (apparently in 1989). I have no other record of his work at the SBG. See under "snails" for a discussion of the aquatic snails of South Creek.

Observation in the SBG forests reveal two groups of invertebrates, millipedes and land snails, both of which are important ecosystem components and enter into the food chains of the forest.

Millipedes

Millipedes are important, mincing up dead leaves and other detritus on the forest floor and passing it through their digestive tract to eventually feed countless bacteria and fungi which sequester nutrients in their tissues until they too die, passing them to the soil where they are reabsorbed by plant roots and associated mycorrhizal fungi. The longer a nutrient such as calcium can be kept in animal or plant tissue, microbial or fungal, the less likely it is to be lost to percolation of rain water.

Because they are so common in the leaf litter and would afford a ready meal for toads, birds, and shrews, slow-moving millipedes have developed chemical defenses. The red and yellow markings on the flat-backed Polydesmid millipedes such as *Sigmoria* and *Cherokia* are warning colorations (Fig. 3). If molested they generate a vesicating secretion of hydrogen-cyanide and smell like bitter almonds (does not damage human skin). Remarkably, this toxic acid is not stored but created at the moment of ejection by two harmlessly stored chemicals in two- chambered defense glands. This is an example of Mullerian mimicry, where a number of species benefit by all having the same warning coloration; it is selected for by predators who learn to avoid the colored individuals. Round millipedes such as *Cambala* lack the flange-like paranota on each side of the body segments and are not protectively colored. Nevertheless, they have single-chambered defense glands and secrete a variety of repellents such as phenols and terpinoids (Roland Shelley, pers. comm.). One is reminded of the colored spots and white markings on some salamanders which have developed defensive slimes. (There are carabid ground beetles that actually spray a noxious chemical.)

Millipedes are not obvious except on really humid days or after rains. Most inhabit the leaf litter. We found eight species (Fig. 4) that fell into our traps or were captured crawling on vegetation or leaf litter. At the SBG you are most likely to see the large *Sigmoria,* the largest of which is quite attractive with pink lateral flanges along the sides. Another common millipede at SBG *(Cherokia georgiana)* is also attractive, appearing to have three rows of yellow "spots." These flat forms like *Sigmoria* are distinct from three, round cylindrical species, the most common of which is *Cambala annulata,* a black and rough-looking form. Oddly, the very large cigarette-sized cylindrical and widely distributed *Narceus americanus* (common at Elachee) was not found at SBG. Interestingly, the hard-shelled *Cambala* is an omnivore, being sometimes carnivorous. While *Sigmoria rileyi* (with red sides) is a foothills and Piedmont form, the SBG *Sigmoria disjuncta* Shelley represents a new county record, a range extension from Hall and Stephens Counties.

Millipedes have two curved antennae and two pairs of legs per body segment. The fast-running centipedes have one pair of legs per body segment. Since they somewhat resemble millipedes they are included here. There may be several species of these leaf-mould carnivores at SBG. The most common is about 40mm long and reddish in color, probably *Scotopochriptops*.The jaws are prominent. There is a small scorpion in the leaf litter fauna *(Vaejovis carolinianus)* which is a predator.

Snails

Land snails seem to occupy more niches than millipedes. Some graze live leaves, some scrape algae, some are detritivores, some eat fungi and a few are active predators. Like aquatic crayfish, snails act as a calcium "sink," thus protecting this vital nutrient from removal by rainwater. On their demise, calcium is restored to the leaf litter and perhaps the soil. Most land snails are "pulmonate" breathing with a vascular "lung" instead of gills. They serve as important food for an array of vertebrates: frogs, salamanders, turtles, mice, shrews, moles and even squirrels and birds. They prefer a moist environment and need calcium for their shells. Most species are small and dwell in the leaf litter and thus are seldom seen. Those we list for the SBG are those which crawl about actively. The largest of these and the most frequently encountered snail at the SBG is the large (35-40 mm) Andrew's snail *(Mesodon andrewsae).* See Fig. 5 for a simple "key" to seven species at the SBG. Fairly common is a similar but somewhat smaller species, the white-lipped forest snail *(Mesodon thyroidus).* If you scan the vegetation in moist areas you can sometimes find three medium-size (15-22 mm) snail species: the strongly ribbed *Triodopsis carolinensis,* one I call the "zebra snail" *(Anguispira alternata)* and the widespread predatory cannibal snail *(Haplotrema concavum)* which is comparatively flat in side view. I did not determine population density of land snails at SBG. However, snails in a hardwood watershed (yVS 18) at Coweeta had a population density of $0.3/m^2$ over soil having 62-76 ppm calcium (Ronald S. Caldwell, Pers. Com.) equal to SBG soils on heath bluffs and dry oak-hickory sites.

As you walk along the Orange Trail up South Creek in the shoaly, rocky area you can find little black aquatic snails of the widespread (83 species) genus *Elimia (Goniobasis).* Their shell is closed by a horny plate or operculum and the small end of the shell is often broken off.

No clams have been found at SBG. Fortunately, the now widespread Asiatic clam was not found in SBG streams but occurs in the streams at Elachee.

Similar species to *Elimia* are intermediate hosts of parasitic worms, especially the blood fluke, *Schistosoma,* the cause of bilharzia or schistosomiasis. The larvae leave the snails and penetrate human skin. (Swimmer's itch in Florida lakes is caused by a similar non-pathogenic species.) In parts of Japan you would not dare wade in brooks such as South Creek.

Most land snails are edible but cook them well. One of the land snails at SBG, *Anguispira alternata,* along with *Zonitoides arboreus,* is implicated in the spread of lungworms in sheep who ingest the snail while eating grass. Land snails such as *Lionella lubrica* are vectors of liver flukes in cattle, deer and hogs. Part of the normal life cycle of the human intestinal fluke *(Fasciolopsis buskiJ)* is in an aquatic snail. An abnormal life cycle involving the liver is said by Clark (1993) to be a principal cause of human cancer. Although she devotes an entire book to it, her theories have not been accepted by the medical community, to say the least.

Earthworms

Earthworms were often abundant in our traps. A spot check of our earthworms by Paul Hendrix revealed that most belong to two exotic species, the European Lumbricus rubellus and the Asiatic Amynthas sp. Both of these are surface-active worms and easily trapped. On the other hand, the native species, such as the genus Diplocardia, are common only below the surface and best captured by digging, hence are rare in our samples. The role of all the exotic worms in the food web of the forest litter is unknown but most considerably augment the food supply of animals such as salamanders and shrews.

VERTEBRATES

Amphibia

Salamanders are secretive under stones and logs and in underground runways and rootholes, foraging on the surface at night. Their numbers are easily underestimated. They are important producers of protein-rich animal tissue. In a northern hardwood forest (Hubbard Brook, New Hampshire) just one species *(Plethodon cinereus)* made four times more tissue than the entire bird community and their biomass was larger than that of the breeding birds in this northern hardwood forest.

Salamanders feed heavily on the litter and soil fauna, in particular worms, larvae, ants, moths, bees, fly larvae, small beetles and probably small crickets. They may even exert some control over populations of some detritivores, thus slowing the rate of litter recycling. They must provide food for some vertebrates as well. Some have developed effective defenses. The common slimy salamander secretes copious quantities of a sticky mucous (difficult to remove from your hands) to deter predators such as shrews. Some *Ambystoma* may use similar tactics (and usually have warning patterns or spots). The genera *Eurycea* and the amphibious *Desmognaths* seem to depend on their agility and speed to disappear in leaves or crevices, or under rocks.

By far the most common salamander at SBG is the slimy (see Appendix 6 for latin names) found under logs in the forest, occurring in all environments except the hydric floodplain. Next in abundance is the three-lined, living in moist environments (except the hydric floodplain). The two-lined species is relatively uncommon (adults become aquatic when breeding). The slow moving marbled salamander *(Ambystoma opacum)* is restricted to floodplains. Immatures emerge in May (they lack marbled markings) and adults are commonly taken in fall and winter. Salamanders are very cold tolerant. The marbled and spotted breed in cold weather (and may be seen swimming under the ice of floodplain sloughs). Hall (1989) records the spring salamander from the "Orange Trail stream" but we did not find it. It occurs elsewhere in Clarke County.

American toads are very abundant at SBG. Very young individuals were seen at least three times between spring and fall, suggesting several breedings. Fowler's toad was taken only on the hydric floodplain. Only one narrowmouth toad was taken (along North Creek). Leopard frogs are relatively common along all the branches. Green frogs do occur. One large adult from the Ivywood beaver pond was tentatively identified as the bronze frog (subspecies *Rana clamitans clamitans)*, a common Coastal Plain form. One definite Coastal Plain form is the green tree frog, which forms a large breeding chorus on summer nights at the Ivywood beaver pond, along with the bass of the bullfrogs and the beautiful whistles of the bird-voiced tree frog, also a Coastal Plain species. These last three species prefer to breed in more or less permanent water (river backwaters, sloughs, etc.).

The green tree frog is partial to freshwater marshes. In the Piedmont, these are created largely by beavers. Except for one specimen (s. Fork of the Broad River) the SBG specimens are the fartherest north records of the green tree frog for the state. According to Williamson and Moulis (1994) this represents the most northerly record on the Oconee River. One other record at Chamblee (DeKalb County) is at about the same latitude. It is unusual to find either of these two Coastal Plain species in the Piedmont. While the bird-voiced is not circumscribed by breeding sites, the green tree frog is. Since the only natural marshes in the Piedmont are beaver-made, one can postulate that beaver are largely responsible for the presence of the green tree frog in the Piedmont. While we have no specimen, I believe that we heard the upland chorus frog, *Pseudacris triseriata feriarum,* breeding in the floodplain below the Callaway Building in February of 1997.

Reptilia

Among the lizards (Appendix 6), the green anole is the most frequently seen, although the five-lined skink is fairly common. An uncommon species is the small ground skink--only one was taken (near the west bluff). Two lizards that inhabit open and successional vegetation, the southern fence lizard *(Sceloporus)* and the six-lined racerunner *(Cnemidophorus)* were not observed or trapped.

Snakes are uncommon at SBG. The southern ringneck is most frequently seen. We found only one each of the other species listed in Appendix 6. With all the toads and frogs, why garter snakes and hognose are not more obvious is puzzling. During the study a watersnake regularly basked in a tree overhanging the beaver dam at the Orange Trail bridge, probably the midland water snake

(Nerodia sipedon pleuralis). Only one venomous snake, a copperhead, was seen crossing the road near the old terminus of the Orange Trail. The black rat snake and kingsnake are inhabitants of the mesic

deciduous forest. Further observation may yield such open field and successional species as the black racer and coachwhip.

Eastern painted turtles are common at the Ivywood Pond. Near the beaver dam of this pond, large turtles are numerous, basking on floating logs. Identification with field glasses of neck markings is not positive. The two best candidates would be the yellow-bellied *(Chrysemys scripta scripta)* and the river cooter *(Chrysemys concinna concinna).* Both of these were taken at the Alcovy (an Ocmulgee tributary) in Newton County, Georgia (Wharton et aI1981). Ivywood Pond specimens need to be collected and identified--a Coastal Plain species is possible. The box turtle, of course, is omnipresent as SBG.

Birds

Birds are one of the few life forms that can always be seen (or heard). You don't have to roll logs or go out at night. A reasonable number of species can be correlated with a given natural environment, ecosystem or plant community.

Appendix 7 lists the common resident summer birds of the SBG, where they nest and where they feed. Fig. 6 is prepared from this list to assist you in determining where certain birds are most likely to occur. Note that 11 species occur over the entire forested area, both floodplain and upland. When uplands are denuded of hardwood forests some of these species are forced into any remaining hardwoods forest, such as may occupy the floodplain. The "green belt" of floodplains thus serves as a refuge for some Piedmont species. Some of these 11 birds, such as the Carolina chickadee, the tufted titmouse and the blue jay could be called "generalists" since they are not too habitat specific. Appendix 7 and Fig. 6 do not cover the Ivywood beaver pond, a more or less permanent marsh unlike anything east of the Oconee. Here we might expect to find additional species such as the great blue heron, red-winged blackbird, perhaps an occasional egret, possibly even rails and bitterns.

Pine forests are grown-up fields, much more extensive now than in pre- Caucasian times. Agricultural or pasture lands are "new" or novel Piedmont environments. Here we find birds seldom seen elsewhere: grasshopper sparrows, meadowlarks, dickcissels, and in the winter water pipets and horned larks who have expanded their original winter range. Vultures, marsh hawks and sparrow hawks have "new" hunting grounds.

To assist you in understanding and identifying birds of the SBG, Jim Richardson, both an ecologist and an ornithologist, has kindly provided me with a written account of a bird walk along the Orange Trail, the levee, and into the powerline and around the buildings. His presentation is so educational and delightful that I include it verbatim as Appendix 8.

Small mammals

Appendix 6 lists five species of small mammals occurring at SBG. If numbers captured is any indication, the tiny southeastern shrew appears to be the most common, followed by the short-tailed shrew. This might be expected in view of the numbers of worms, beetles, snails, camel crickets and other food available. The short-tailed shrew is known to store underground caches of live snails paralyzed by the venomous secretions of certain salivary glands. They are thus one of the few venomous mammals in the world. The least shrew prefers successional areas-- we took only one or two specimens.

Among the rodents, both the pine vole and white-footed mouse are present but, I would guess, not very abundant, at least between May and September. Since rodents are notorious for population cycles, we may have experienced a "low" period. I do not recall having seen a chipmunk at the SBG. Rice rats *(Oryzomys palustris)* might be expected at the Ivywood Pond.

Larger mammals

Little effort was made to document the larger and more mobile mammals.

Deer are very common. They and domestic hogs and cattle may have had a profound influence on some herbaceous species of plants and tree seedlings. Although a few deer appear to be crossing the deer fence, it remains to be seen if there will be a change in the plant life within the fence boundary. Along the small branches, tracks of raccoon and mink have been seen. Along the Oconee, beaver are present, possibly muskrat and an occasional otter could be expected. We can reasonably expect skunk, opossum, grey squirrel, fox and bobcat to be present. Coyotes may appear from time to time.

HOW DO WE TIE IT ALL TOGETHER?

(As a final "chapter" and to help balance all the data, I include a lecture I gave to a large group of science teachers at the Callaway Building, June 25,1996.)

I was fortunate recently to be invited to accompany your group of 40 schoolteachers to Whiteside Mountain and the Wolf Knob Cove. I heard two scientists explain some of nature's complexity to you. Giles Allard described the rock types and Joshua Laerm talked about the diversity of vertebrate life in the mountain forests, emphasizing the variety of tiny, carnivorous shrews. In view of your dedication to teach young minds who stand to inherit the consequences of the use and abuse of our planet, I'd like to try and relate these two viewpoints.

The Bible tells us that, in the beginning, the Lord said, "Let there be light." Science has since verified that from this first primordial event there was indeed light,and from this event, all else resulted. Some call it the "big bang," or it may have been a supernova in "our" corner of the universe. Either way the most visible manifestation was the incredible speed and glory of light. Whether originating from tiny masses of infinite density called "singularities," or from a small fireball containing the total mass of the universe, or from the more localized explosion of a collapsed giant star called a supernova, "matter" as we know it started forming from practically nothing, from a soup of unimaginably small neutrons, protons and energy. The tiniest "first" atom, hydrogen, is as near nothing as we can imagine. However, two of them soon fused to form helium, the second element and twice as heavy. This reaction produced enormous heat and light. In this stellar furnace protons were added to protons, and neutrons were converted to protons, so that with each nuclear fusion a new and heavier element formed. Whether done in low-mass stars that retained their iron core or in gigantic high-mass stars that collapsed to hurl their matter into space, all 90 plus elements comprising the building blocks of the known universe were formed. The heavier elements gravitated together to form planets like the earth. The stuff of life was thus born in the alchemy of stars.

While some 100 plus elements exist, only less than a quarter of them are essential to living things. The commonest of these, many of them metals, occur abundantly in molten rocks deep within the earth and hardened on its surface. This rock I hold in my hand, the gneiss from Whiteside Mountain, when ground up, will yield about 22 of these heavier elements formed in the fiery cauldrons of space. If you add elements now in gaseous form in the atmosphere (nitrogen, carbon) and from the rain (iodine, sulfur) you have the ingredients to build every plant and animal that lives or has ever lived on this earth, from the tiniest bacterium to the great dinosaurs. They are made from nothing more or less than the rock that I hold in my hand and from the liquids and gases associated with either the formation or disintegration of rocks or from incoming asteroids and comets. From dust thou art truly descended.

If we grind up this rock I am holding, or dissolve it as nature does, we still do not have one of terrestrial life's most essential components--soil. Some simple life forms can "feed" on the energy of chemical compounds alone, usually in water such as around a thermal deep sea vent. But, aside from some parasitic forms, most green plants on land cannot grow without soil. Soil not only contains nutrients in the rock from which it was derived, but it houses many life forms, from bacteria to millipedes, that convert dead plants back into usable food molecules.

On land at least, we cannot have animals without plants. For example, most terrestrial animals must get the element nitrogen from either living plants or from their remains in the mantle of soil covering this great ball or rock we call the earth. This mantle that conceals the bedrock of mountains like Whiteside is but a thin layer of weathered rock particles, converted into true soil by plants and its contained host of detritus-eaters. If you want to see what Whiteside would look like without its thin soil covering blanket, take a look at Stone Mountain near Atlanta. Plants not only aid in making soil, they must protect and anchor the precious medium that perpetuates their own growth and future. This they do with their leaves, twigs and roots. Otherwise the thin covering that means life to them would be washed and blown away, leaving but naked stone against the sky.

Countless growing seasons must come and go. Each year a few more specks of decayed dead leaves and twigs are added to the minerals of the forest floor. In unglamorous terms we call this stuff of decayed life "organic matter." This fertile layer, darkened by organic material, we call topsoil. This name hardly does justice to what is very possibly the most precious natural resource on earth. Teeming with microscopic life, it is the basis of all food chains that support the wealth and health of nations of animals and peoples alike. Unfortunately, mankind is slow in realizing how terribly vulnerable topsoil is. Stripped of its protective plant cover by deforestation, monoculture crops or overgrazing, the accumulation of millennia is carried away by the waters and the wind. Most nations have neither the time nor the money to restore it.

Only a handful of human cultures, by carefully tending rice paddies or with the aid of terraced agriculture, have been able to develop a sustainable yield of food plants from soils which once supported natural plant communities. Our own society, by continuing to exploit and export as corn or wheat the deeper topsoils of the mid-west prairie states or the peats of the Everglades, and by using wealth derived from fossil fuels, is able to practice a stop-gap form of agriculture with artificial fertilizers. But, in so doing, we are violating a natural"law"--the system by which life as has built and maintained itself upon the naked rock.

If our national budgets can stand the strain we can continue not playing by nature's rules, for a while at least. We can purify air at great costs. If we remove all the forests from the earth and our wells and streams go dry because of it, we can always distill sea water--again at great costs. But once the little skim of topsoil disappears, it is gone forever for all practical purposes. If we abandon a once forested field, it will slowly revert to climax forest again in maybe 100 years. If we are lucky one inch of topsoil will have formed. At best, our virgin Piedmont soils had perhaps 12 inches of topsoil. By the time the boll weevil closed the cotton era in the 1930's, erosion from the practice of row-crop agriculture had stripped all the topsoil from most of the Georgia Piedmont. If any topsoil remains, we owe it to slopes too steep to plow.

So we have a sequence of events, initiated by the Creator when He flooded the galactic systems with light, culminating, for us at least, in the sun, rocks, water, soil, air, plants and animals.

Native Americans revered and respected these things. They knew all were important to give them life. To them, even rocks had being and deserved appropriate respect. When they dug a plant, such as ginseng, they left a pinch of tobacco in exchange, or prior to killing a deer, they asked the animal's spirit to forgive and understand the taking of its life for food. Today, science is close to the ethic of the native Americans. The modern concept of the ecosystem requires that we understand and value all respects of it, living and non-living, before we can have a functional unit of nature. Such "units" give our natural environment enormous diversity and resilience that make our own lives both exciting and, above all, sustainable.

There you have it. No source of light, no rocks (and nothing for Dr. Allard to explain); no rocks, no soil or plants, at least on land; no plants, no oxygen and no animals--not a single salamander or shrew and, for that matter, no Dr. Laerm himself. And so we come full round.

Even as apex omnivores with large brains and an enormous store of accumulated knowledge we continue, with child-like curiosity, our search for a "theory of everything." Our active minds embrace larger and larger "pictures" of how it all came to be. According to the ecotheologian Thomas Berry, the creation story must be expanded. It must include the origin of light and those five species of the shrews threading through the leaf litter of a mountain forest and beyond. We have been allowed to develop the wisdom and tools to fill in the details and time-scale of how things came about.

The new creation story we must learn is even more awe-inspiring than . anything from the past. With every breath, we breathe molecules exhaled by dinosaurs. Like many of the eastern faiths before us we, and "we" includes many of our leading physicists, are coming to realize that we are indeed with all life if not with the universe itself. Along with this realization has come the unsettling truth that, with all our ancient knowledge, our technology and our sheer numbers, we have placed the air, the water and soils, and thus ourselves at peril. If we reject a new creation story, the refuge of an afterlife will become increasingly attractive.

Unlike Native Americans and some Eastern faiths, the Christian religions, emphasizing God-man and man-man relationships, have delayed the adoption of an ethic involving God, man and the land, a relationship that is imperative for our earthly survival. Given the way human history is taught, grounded in the Greek philosophy which recognized a dualism between matter and spirit, it is not easy for us to accept our obligatory role as an ecosystem member, either by intellectual or spiritual unity. In the past our numbers were few and it did not matter so much--we could always migrate to other lands, exploit "virgin" soils beneath former forests or wetlands or, with once-plentiful water, make a desert bloom. Now, survival means understanding how nature works in all the bioregions of the earth, not just the one we live in, for our technological octopus feeds from every corner of the globe.

It is important for us as teachers, to get people, both young and old, to have a direct experience with nature in natural environments, the forests, prairies, rivers and their flood plains, swamplands and sea shores. In the experience we have just had here in the Blue Ridge I hope that we have shared some understanding of the oneness of all things or "how things tie together." A virtual reality experience by video can be valuable, and computer webs can be adjunct learning sites. But they are counter-productive if they divert our minds from the mutually consistent and eternal relationships of the real web out there-the events and processes that unify all things in an ecological, earth-oriented spirituality.

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APPENDICES

Appendix 1 Calcium, magnesium, pH, and CEC of soil samples collected at the SBG.

(If two samples, depths are 6" and 12", if three the third sample is at 24". Horizontal lines separate the 37 sites).

SAMPLE NO.	CALCIUM (lbs/A)	MAGNESIUM (lbs/A)	pH	CEC (meq/100g)
1	97	136	5.5	4.2
2	72	341	5.0	5.5
3	75	42	5.1	4.1
4	83	53	5.1	3.7
5	35	14	5.0	2.2
6	54	149	5.3	3.6
7	67	33	4.9	3.6
8	95	186	5.3	4.0
9	98	76	5.1	4.3
10	71	106	5.3	3.5
11	337	164	5.5	3.7
12	188	203	5.5	3.4
13	95	55	5.1	2.9
14	49	13	4.9	2.6
15	589	165	5.6	5.9
16	272	398	5.5	6.5
17	376	102	5.3	5.1
19	78	90	5.2	4.2
20	68	128	5.3	4.0
21	55	354	5.3	5.2
22	160	159	5.0	5.1
23	122	224	5.0	4.9
24	144	291	5.2	5.3
25	809	151	5.9	6.0
26	341	92	5.3	4.6
27	297	145	5.2	5.7
28	1689	199	6.0	10.5
29	367	75	5.3	5.4
30	231	214	5.6	4.9

SAMPLE NO.	CALCIUM (lbs/A)	MAGNESIUM (lbs/A)	pH	CEC (meq/100g)
31	767	234	6.1	5.9
32	384	254	5.6	5.4
33	194	173	6.0	3.9
34	116	241	5.5	4.2
35	278	90	5.6	3.6
36	247	124	6.0	2.4
37	373	231	5.4	5.2
38	266	320	5.5	5.7
39	109	64	5.3	4.1
40	191	265	5.6	5.3
41	1611	405	5.7	9.5
42	756	439	5.7	7.4
43	697	201	6.4	5.6
44	330	199	5.4	6.7
45	948	436	6.2	8.9
46	424	349	5.8	5.8
47	321	109	5.4	4.6
48	234	133	5.4	4.5
49	296	128	5.7	3.8
50	193	152	5.4	4.1
51	206	130	5.3	4.4
52	77	263	5.2	4.6
53	155	155	5.6	4.3
54	101	156	5.5	5.0
55	73	86	5.2	3.8
56	81	78	5.3	3.4
57	181	151	5.2	4.1
58	185	222	5.4	4.4
59	81	132	5.3	4.4
60	66	139	5.5	4.5
61	324	228	6.0	5.7
62	213	235	5.7	4.8

SAMPLE NO.	CALCIUM (lbs/A)	MAGNESIUM (lbs/A)	pH	CEC (meq/100g)
63	60	60	5.2	3.6
64	41	76	5.1	4.8
63A	637	178	5.7	6.5
64A	385	151	5.6	5.4
65	328	250	5.7	4.8
66	216	264	5.5	4.2
67	305	137	5.4	4.7
68	221	162	5.6	4.5
69	244	137	5.2	5.7
70	169	149	5.4	5.2
73	98	55	5.1	4.6
74	115	121	5.3	4.2
75	190	87	5.3	4.1
76	118	77	5.3	3.5
77	111	55	5.1	4.2

Appendix 2. Station numbers with field notes of interest

(see Map 1 for location of each station)

1. gate to trail across floodplain
2. green ash-box elder canopy
3. green ash, 36 DBH (diameter at breast height)
4. sweetgum -- mulberry (probably on levee)
5. grove of hickories and one sweetgum on Congaree soil
7. river birch in low swale fringed by sick loblolly pines on Inner edge of floodplain
8. begins mature oak-hickory
10. down old terrace
14. large scarlet oak
15. timber young, in succession - young shagbark hickory
16. drier aspect with white oak, southern red oak
17. Kingsnake observed
18. gate on Green Trail
19. fly poison below trail, fringe tree
20. devil's bit in bloom (May 4)
22. marsh-canopy green ash, river birch, red maple
26. pines from old agriculture visible north of branch
34. huge privet patch .
41. galax
42. featherbells, dog hobble
45. second growth forest
46. soil profile in creekbank, 48" down to saprolite above paragneiss
47. many stumps of large pines
48. erosion gullies, large mature loblolly pines
49. netted chain fern
51. big ash, big bitternut hickory (24" DBH)

52. 55. Michaux's lily

59. at "dead coon spring" large white oak shows soil profile with gray (ferrous iron) soil (anaerobic conditions) with adjacent red (ferric iron) soil (aerobic conditions)
60. dense Christmas fern
61. erosion gullies, a few large pine remain
63. trees are mature
64. yellow patch
65. American chestnut
66. heavy Christmas fern, hepatica, lovage on north facing ravine wall. Good wildflower site.
67. Beech - hop hornbeam "flat"
68. soil profile exposed in creek bank
69. maiden hair fern
72. beech-dominant forest
73. mature sweet pignut - pignut hybrids

73. 74. fence runs E-W - patches of pokeberry occasional

75. rock pile at fence corner
76. storax (?)

77.78. watch for Canadian avens, thimbleweed, coral honeysuckle, bellworts (2 sp.), dwarf pawpaw, agrimony, sapling cucumber magnolia, wild onion, wild yam, etc.

79. Georgia hackberry, buckthorn burnelia, an exotic elm (Ulmus parvifoJia)

<u>Through gate and across floodplain (</u>note: Tall composites (crown beard, wingstem), cinnamon vine, passion flower, vine mint)

85 - 50 - 52. N. end of bluff pallid hickory, Georgia mint, Michaux's lily, Oconee azalea
88. Purple Trail - snakeroot, upland possum haw
89. dry mesic forest, (no white oak), black oak, scarlet oak
93. silverbell, sweet shrub, dwarf buckeye, mulberry
97. this bluff seems to lack galax, Georgia mint, hawk weed
96. ring neck snake under rocks
97. stand of shortleaf pine in deciduous forest
98. stand of large mature black and N. red oak (30" DB H)
99. old cut pine stumps indicating logging
100. springhead, bog area, 1st *Desmognathus* salamanders! note crayfish "chimneys"
101. large black cherry (12" DBH) even with corner chain length fence
102. huge serviceberry (3 trunks ea 6" DB H)
103. beech-dominated ravine forest
104. beaver pond ecotone - silky dogwood, elderberry, willow, ground nut vine, virgin's bower vine, beadfern, lizard tail (blooms May 24)
105. painted turtle caught, huge loblolly pine (30" DBH)
106. 1 and 3 lined salamanders
107. begin beech dominant forest
108. erosion gullies from agriCUlture upslope, some 60 year old yellow poplar
111. big loblolly pines to the east (in succession)
113. in succession back to hardwood forest but trees tiny (to west); goes into white oak/hickory approx. 50 years old
116. white oak dominant mature (50 + years)
120. devil's bit at south end of Yellow Trail bridge
122. mature forest, white oak 2.3 m CBH (circumference breast height), N. red oak (labelled) dia. 3.15 m DBH
123. large privet patch with winged elm
125. patch of pale Indian plantain
130. large switch canes in a brake, creek is relatively unpolluted (South Creek)
132. from first falls on South Creek up stream geology is rewarding - paragneiss and orthogneiss and amphibolite lenses form falls in branch. Complex layering and folding of the highly metamorphosed rocks is visible. Dark mafic amphibolite outcrops in Orange Trail 20' south of blazed ironwood tree.
141. Bladdernut (calciphile) patch of 15+ plants
142. nice white oak, N. red oak grove (mature trees)
143. worm snake taken under log
146. amphibolite "float" (surface rocks) in trail near here
147. dense subcanopy chalk maple with red bud seedlings (piece of amphibolite rock found at 10" depth in soil)
150. rock outcrop with bladdernut patch
151. horsesugar patch
153. Amorpha, 48" black ratsnake observed Aug. 17, 1996
154. heavy redbud patch, bloodroot, geranium; soil high in calcium (soil sample #25), dark red- brown
155. redbud - mulberry over high calcium soil (pH 6.0, soil sample #28)
156. remarkable number of sapling sweetgums in understudy of evidently moist, rich ravine
157. soil sample #33
160. bad sheet erosion from field
163-164. chalk maple (5" OBH) steep bluff (pH meter reads 6.9)
166. line where dark amphibolite rock "strikes" across amphibolite branch. Large pieces are visible.
171. large outcrop of paragneiss on north face of ravine
172. large black haw by Purple Trail
172A. beaver pond on South Creek. (0) green ash, (0) sycamore, (0) river birch, (0) American elm, (0) sweet gum -- After June 15 pond becomes covered with duckweed, a tiny flowering plant.
173. laurel oak east of gate next to a persimmon and a scarlet oak
174. some amphibolite float in White Trail

178. chalk maple 70' below lowest terrace of old cotton fields
183. impressive mature oak-hickory-maple forest
184. fine mature N. red oak-white oak forest
185. mature N. red oak-white oak forest with black cherry, dogwood and muscadine
187. not in succession but has had livestock impact (hog wire enclosures, etc.) timber mature, with scattered big Southern red oaks.
189. shagbark hickory, much young chalk maple and red bud
190. huge chalk maple (6" OBH), densest Christmas fern on property
191. excellent stand of large oaks (N red, black) with silverbe", chalk maple and maple-leaf viburnum, some red bud, and dwarf paw paw. No hop hornbeam this community.
192. giant black oak (36" OBH)
195. old pine stumps in this area
198. area with shagbark hickory
200. at gate is post oak-shagbark hickory-mulberry-red bud community

Appendix 3   Plants of the SBG (separate document)

Appendix 4 Insects of the SBG

BEETLES *(Coleoptera)*

- Long-horned Beetles *(Cerambycidae)*
  - *Prionus imbricornis (I.)*
  - *Orthosoma brunneum* (Forster)
- Wood Beetle *(Disteniidae)*
  - *Distenia undata* (Fabr.)
- Bessbugs *(Passalidae)*
  - *Popilius disjunctus* (IIliger)
- Scarabs *(Scarabaeidae)*
  - *Deltochilum gibbosum* (Fabr.)
  - *Xyloryctes jamaicensis* (Drury)
  - *Pachystethus marginata* (Fabr.)
  - *Geotrupes spJendidus* (Fabr.)
  - *Geotrupes egeriei* (Germar)
  - *Trox variolatus* (Melsheimer)
  - *Copris minutus* (Drury)
  - *Glaphyrocanthon viridis* (Beauvios)
  - *Onthophagus hecate hecate* (Panzer)
- Snout Beetles *(Curculionidae)*
  - *Panscopus impressus* (Pierce)
- Round Fungus Beetles *(Leiodidae)*
  - (?) *Neocyrtusa* sp.
- Sap Beetles *(Nitidulidae)*
  - *Stelidota geminata* (Say)
- Carrion Beetles *(Silphidae)*
  - *Nicrophorus orbicollis* (Say)
- Ground Beetles *(Carabidae)*
  - *Dicaelus dilatatus* (Say)
  - *Dicaelus* sp.
  - *Galeritula janus* (Fabr.)
  - *Scarites subterraneus* (Fabr.)
  - *Chlaenius* sp. (?) *aestivus* (Say)
  - *Pasimachus* sp. *(depressus / punctulatus)*
  - *Agonum punctiforme* (Say)
  - *Pterostichus* sp.
  - *Evarthrus sigillatus* (Say)
  - *Harpalus* sp.
- Click Beetles *(Elateridae)*
  - *Hemicrepidius memnonius* (Herbst)
- Rove Beetles *(Staphylinidae)*
  - *Platydracus maculosus* (Gravenhorst)

TRUE BUGS (*Hemiptera: Heteroptera*)

- Assassin Bugs *(Reduviidae)*
  - *Rhiginia cruciata* (Say)

ANTS & RELATIVES (Hymenoptera)

- Ants *(Formicidae)*
  - *Camponotus* sp.

CRICKETS & GRASSHOPPERS (Orthoptera)

- Shield-backed Grasshoppers *(Tettigoniidae)*
  - *Atlanticus* sp. (?) *gibbosus* (Scudder)
- Camel Crickets *(Gryllacrididae)*
  - *Ceuthophilus* sp. 1 [prob. *gracilipes* (Haldeman))
  - *Ceuthophilus* sp. 2
  - *Ceuthophilus* sp. 3

Scorpions *(Arachnida; Scorpionida; Vaejovidae)*

- Vaejovis carolinianus (Beauvois)

Appendix 5 Common animals in Orange Trail stream

(Identified by Bob Hall)

Mayflies (Ephemeroptera)
    Baetidae *Baetis*
    Heptageniidae *Stenonema*
Plecoptera (Stoneflies)
    Peltoperlidae
    Tallaperla maria (This is the type location for this species. At least three other families in here that I cannot remember.)
Dragonflies and Damselflies (Odonata)
    Cordulegastridae *Cordulegaster*
    Calopterygidae *Calopteryx*
Bugs (Hemiptera: Heteroptera)
    Gerridae
Megaloptera (Dobsonflies)
    Corydalidae *Corydalus* (Hellgrammite)
Trichoptera (Caddisflies)
    (Probably lots of other families; I have not keyed some of those below to genus.)
    Hydropsychidae
    Limnephilidae
    Philopotamidae
    Rhyacophilidae *Rhyacophila*
Diptera (Flies)
    Chironomidae (Midges)
    Tipulidae (Crane flies)
    Simuliidae (Black flies)
Coleoptera (Beetles)
    Ptilodactylidae *Anchytarsus*
Decapoda (Crayfish - not keyed to genus)
Gastropoda - Prosobranch (gilled) snails
Salamanders
    Desmognathus spp.
    Gyrinophilus (Spring salamander)

Appendix 6 Amphibians, Reptiles and small mammals of the SBG

Amphibians

Common Name	Name in original report	Current Scientific Name
Spring salamander *	*Gyrinophilus porphyriticus*	*Gyrinophilus porphyriticus*
Spotted dusky salamander	*Desmognathus fuscus conanti*	*Desmognathus conanti*
Southern two-lined salamander	*Eurycea (bislineata) cirrigera*	*Eurycea cirrigera*
Three-lined salamander	*Eurycea longicauda guttolineata*	*Eurycea guttolineata*
Slimy salamander	*Plethodon glutinosus*	*Plethodon glutinosus*
Marbled salamander	*Ambystoma opacum*	*Ambystoma opacum*
Bird-voiced tree frog	*Hyla avivoca*	*Hyla avivoca*
Grey tree frog	*Hyla (versicolor) chrysocelis*	*Hyla chrysocelis*
Green tree frog	*Hyla cinerea*	*Hyla cinerea*
**Upland Chorus frog		*Pseudacris feriarum*
***Spring Peeper	*Hyla crucifer*	*Pseudacris crucifer*
**Cricket frog		*Acris crepitans*
Southern leopard frog	*Rana sphenocephala*	*Lithobates sphenocephalus*
Green frog	*Rana clamitans*	*Lithobates clamitans*
Bull frog	*Rana catesbeiana*	*Lithobates catesbianus*
American toad	*Buto americanus*	*Anaxyrus americanus*
Fowler's toad	*Buto woodhousii fowleri*	*Anaxyrus fowleri*
Narrow-mouthed toad	*Gastrophryne carolinensis*	*Gastrophryne carolinensis*
**Eastern Spadefoot toad		*Scaphiopus holbrookii*

*(Bob Hall Record) **New entry; not in original ms. ***corrected common name (Cricket frog)

Reptiles

Common Name	Name in original report	Current Scientific Name
Five-lined skink	*Eumeces fasciatus*	*Plestiodon fasciatus*
Ground skink	*Scincella lateralis*	*Scincella lateralis*
Green anole	*Anolis carolinensis*	*Anolis carolinensis*
Northern copperhead	*Agkistrodon contortrix mokasen*	*Agkistrodon contortrix mokasen*
Southern ringneck snake	*Diadophis punctatus punctatus*	*Diadophis punctatus punctatus*
Worm snake	*Carphophis amoenus*	*Carphophis amoenus*
Black rat snake	*Elaphe obsoleta obsoleta*	*Pantherophis obsoleta obsoleta*
Eastern king snake	*Lampropeltis getulus getulus*	*Lampropeltis getula*
Midland banded water snake	*Nerodia sipedon pleuralis*	*Nerodia sipedon pleuralis*
**Plainbelly water snake		*Nerodia erythrogaster*
Box turtle	*Terrapene carolina carolina*	*Terrapene carolina carolina*
Painted turtle	*Chrysemys picta dorsalis*	*Chrysemys picta picta*

** New entry; not in original ms.

Small mammals

Southeastern shrew	*Sorex longirostris*
Short-tailed shrew	*Blarina brevicauda*
Least shrew	*Cryptotis parva*
White-footed mouse	*Peromyscusleucopus*
Pine vole	*Microtus pinetorum*

Appendix 7 Common resident summer birds of the SBG

Species	Nest Info	Foraging info
Green Heron	Platform In trees at river edge	Fish from pools, Oconee River and beaver ponds
Wood Duck	Hollow tree near river or pond	Beaver pond
Red-shouldered Hawk	Platform in large deciduous tree	Snakes, lizards, rodents from all habitats
Mourning Dove	Gardens, fields, power line cut	Same
Yellow-billed Cuckoo	Deciduous forests	Deciduous forests, field edges, power line
Barred Owl	Old hollow tree in river bottom	Rodents from fields, clearings
Whip-poor-will	On ground, dry forest hillsides	Clearings, at night
Chimney Swift	Not In Garden	Foraging in airspace over canopy
Ruby-throated Hummingbird	River corridor	Visits flowers in all habitats
Belted Kingfisher	Burrow In river bank	Fish from river pools, beaver pond
Red-bellied Woodpecker	Cavities in available standing dead trees	All forested habitats
Downy Woodpecker	Cavities in dead snags	All forested habitats
Northern Flicker	Cavities in dead trees	All forested habitats
Acadian Flycatcher	Understory near streams, river	Same
Great Crested Flycatcher	Mature deciduous forest	Same
Eastern Phoebe	Callaway Building	Clearings
Blue Jay	All forested areas	Same
American Crow	All forested areas	All habitats
Carolina Chickadee	Hollow branches, forested areas	All forested habitats
Tufted Titmouse	Hollow branches, forested areas	All forested habitats
Brown-headed Nuthatch	Pines	Mid-successional forest with mature pines
Carolina Wren	All habitats	All habitats
Gnatcatcher	Riverbottom forests	Same
Eastern Bluebird	Bird houses, hollow limbs	Gardens, power line, pastures
Wood Thrush	Ravines, mesic deciduous forest	Same
American Robin	All habitats	All habitats
Mockingbird	Gardens, power line, pastures	Same
Brown Thrasher	Thickets, forest edge, river edge	Same
White-eyed Vireo	Thickets, forest edge, power line	Same
Red-eyed Vireo	Mature deciduous forest	Same
Pine Warbler	Pines	Pines, ear1y successional forest
Prothonotary Warbler	Hollow snags, beaver pond, river	Same
Swainson's Warbler	Privet thickets, river bottom	River bottom, flood plain
Louisiana Waterthrush	Mesic ravines, creek banks	Deciduous understory forest
Hooded Warbler	Deciduous forest, floodplain	Same
Kentucky Warbler	Mesic deciduous forest, ravines	Same
Yellow-breasted Chat	Early successional clearings	Forest edge
Summer Tanager	Pines	Pines, early successional forest
Northern Cardinal	Forest edge, thickets, gardens	Gardens, forest edge, clearings
Indigo Bunting	Forest edge, early successional	Gardens, power line, forest edge
Rufous-sided Towhee	Forest edge, early successional	Gardens, power line, forest edge
Chipping Sparrow	Gardens, power line	Gardens, power line, forest edge
Field Sparrow	Power line, abandoned fields	Same
Eastern Meadowlark	Fields at Garden entrance	Same
Common Grackle	Pines, gardens	Same
Brown-headed Cowbird	Forest edges	Parasite of forest nesting birds
Parula Warbler	Levee forest, riverbottom	Levee, river forest

Appendix 8 A bird walk through the SBG

(Text by Jim Richardson.)

A bird walk at the State Botanical Garden in late May is a lesson in the ecology of natural communities of the Piedmont Region of Georgia. A bird walk offers a window on the history of the Piedmont as well, reflecting anthropogenic changes to the land over the last several hundred years. The composition of bird species observed on a May walk reflects differences that may be noted in the natural environments of the Botanical Garden. This variety of environments may be observed along the Orange Trail, the Oconee River levee, the tailored gardens by the Conservatory, and a multitude of other beautiful locations. We will discuss a few of the more common bird species expected on a bird walk at the Garden.

As we prepare to enter the Orange Trail by the upper parking lot, several common bird species can usually be observed that never nest within the Garden or even alight on its grounds. These are birds of the airspace above the Garden, including chimney swift and barn swallow by day and the nighthawk at dusk. The forest, gardens and fields produce prodigious numbers of insects that disperse on the wing above the canopy, providing an important source of food for birds able to glean the airspace for insect food. Also in the airspace above us are black and turkey vultures cruising hundreds of miles daily across the landscape in search of dead creatures, and red-tailed and red-shouldered hawks. The hawks do nest in the Garden, but they are not often seen except when they rise on the thermals above us in courtship aerobatics or to defend loudly their nesting territory against other hawk pairs.

The head of the Orange Trail enters an old cotton field, although it may look like a forest at first glance. Erosion control contour lines are still in evidence. considerable numbers of pine trees reflect an earlier successional stage that is only now being displaced by sweetgum and a few hardy oaks. The forest floor is open and sunny, with dry leaf litter and little soil. This is home to rufous-sided towhees, brown-headed nuthatches, and pine warblers. It is also home for chuck-will's- widows and whip-poor-wills that are rarely seen by day but often heard at night. If the occasionally seen black-and-white warbler has chosen to nest within the Garden, it will be in this type of dry habitat that you will find it. With a steady supply of dead, standing pine trees, cavity nesters should also be very much in evidence, including red-bellied woodpeckers and flickers. Although common at the trail head, these woodpecker species may also be found feeding elsewhere in the Garden where trees have died recently. They specialize on the larvae of wood- boring beetles.

As the Orange Trail reaches the ravine, a significant change occurs within the forest community. Pines. have all but disappeared, having been excluded by hickory, oak, and tulip poplar, to mention a few of the forest canopy dominants. A small stream is an important element not seen before. Light levels are less than at the trail head, and the humidity is higher. Soils are richer and damp, with ferns and mesophylic herbs. Dogwood, redbud, and ironwood have become predominant components of the understory. Ravine nesting birds are usually noticed by their song before being seen. Their ringing calls seem designed to be heard within the dense confines of the ravine and above the noise of a rushing brook. Kentucky and hooded warblers are common here, but the Kentucky call is often confused with the Carolina wren, also a common resident. Most loud of all is the territorial song of the Louisiana waterthrush near its nest on the bank of the creek, Finally, an explosive squeak is the chosen song of the Acadian flycatcher. This most indistinct of the confusing Empidonax flycatcher group is best distinguished by its strange little song. Check for a nest suspended on a branch positioned low over the water of the stream.

Garden communities are partitioned vertically in space as well as horizontally. Particularly is this true within the ravines where a diversity of tall trees form a dense canopy high above the stream. This important natural community supports an assemblage of common resident birds quite distinct from the understory species mentioned before. Common canopy species of the deciduous forest include the red-eyed vireo, great-crested flycatcher, and yellow-billed cuckoo. All of these birds forage for insects in the canopy and nest there as well. Because of the great height above our heads and the dense foliage, canopy residents are most easily located by their song.

As the Orange Trail reaches the Oconee River floodplain, the forest community changes dramatically, and so do the resident nesting birds. The floodplain extends into the ravine several hundred yards from the river, where sediments are deposited in deep layers during spring floods. Saturated clay soils support cane stands, Chinese privet, river birch, swamp maple and sycamore, while the oaks, tulip poplar, and sweet gum begin to drop out. A *beaver* pond forms a sunny clearing within the forest. Apparently, the *beaver* stimulate the growth of young shoots at ground level by girdling but not felling the inundated trees. Because of this behavior pattern of the *beavers,* woodpecker species are again common, as they were in the dying pines at the trail head. Green herons and wood ducks find ideal foraging habitat at the pond, as does the wood peewee perched on a dead limb *above.*

The ringing call of the prothonotary warbler identifies this common resident of *beaver* ponds. Dead snags provide convenient nest cavities for this brilliant warbler cloaked in prothonotary yellow. Where Chinese privet grows in impenetrable stands, a short detour will often flush a woodcock that may *have* been probing for worms in the damp soil just before we *arrived.*

Within the *privet* thickets we will also find the elusive Swainson's warbler that can be heard for great distances but rarely if ever allows you a glimpse of its cryptic brown plumage. Before we introduced privet from China, the thicket home of the Swainson's was probably *river* cane that has now largely been displaced by the aggressive privet. Man changes the environment, while bird species are required to adapt or perish. The Swainson's warbler is still with us in reduced numbers, but the Bachman's warbler and ivory-billed woodpecker *never* made it.

Arriving at the Oconee River bank, the Orange Trail suddenly rises onto the *levee* which has been molded from deposited sand, the heaviest and therefore first of the floodwater sediments to settle out at the river's edge. Clay particles are the lightest and are therefore carried farthest from the river at flood time. The levee forest community is not dissimilar from the tree species by the beaver pond, but the vertical forest walls on either side of the river form a lush, sunny landscape that is dramatically different. Climbing vines are a rich addition to this diversity and, as we might expect, the bird community again reflects the changes in the forest structure. The ruby-throated hummingbird is more common here, as are several other canopy species like the blue-gray gnatcatcher and the parula warbler. Because sunlight reaches the bank, dense thickets of foliage grow by the river, providing an ideal home for cardinals, towhees, and more of the ubiquitous Carolina wrens. The river itself forms a highway flyway for swallows, grackles, and great-blue herons, to name but a few.

We leave the Orange Trail to continue along the Oconee River levee until the powerline right-of-way. This is a dramatic change in forest community structure from closed floodplain forest to an open gap of significant magnitude. Members of the summer bird community residing along the powerline represent our most common residential members, for these are the species that have adapted to the Piedmont transformed by the hand of man. Common yellowthroats seek the weedy patches that have not been mowed for a month. White-eyed vireos and chipping sparrows call from clusters of dogwood and plum thickets, while indigo buntings and yellow-breasted chats prefer the forest edge here in the right-of-way better than along the river. Bluebirds are resident in boxes provided for them. Right-of-way birds were probably less common before the arrival of man, utilizing occasional clearings resulting from cataclysmic events such as fire or tornado swaths. Right-of-way bird species have thrived and multiplied in our backyards, while old growth canopy species have surely dwindled.

Our walk ends among the formal gardens of the Callaway Building and the Conservatory. Here, we find robins, mockingbirds, and the ubiquitous Carolina wrens. A phoebe uses a convenient Callaway window ledge for its nest site, where once it must have depended on overhanging rock formations. Horticulturally favored pines have been maintained among the gardens, and so the summer tanager and pine warbler add themselves to the species mix. Birds associated with the formal gardens are true backyard species of the Georgia Piedmont. Add to this a newly arrived resident from the far West, the house finch, that survives in its new home by standing close to man and his urban environment rather than entering forested habitats that are foreign to its memory.

From the parking lot, we depart for the Botanical Garden front entrance gate and are treated to one more reminder that bird communities reflect the environment. Surrounding the gate are pastures, a taste of "prairies" where such spaces never existed in abundance before man cleared the Piedmont forest several hundred years ago. Meadowlarks, field sparrows and an occasional dickcissel might best be described as typically "western" birds that have found a home in the Piedmont, thanks to our landscape altering efforts.

Figures

Fig. 1. Possible origin of the rocks of the SBG.

The dominant metamorphic common rock, paragneiss, is thought to be derived from sedimentary rocks whose source was probably volcanic. Injected layers of amphibolite were probably derived directly from melted basalt.

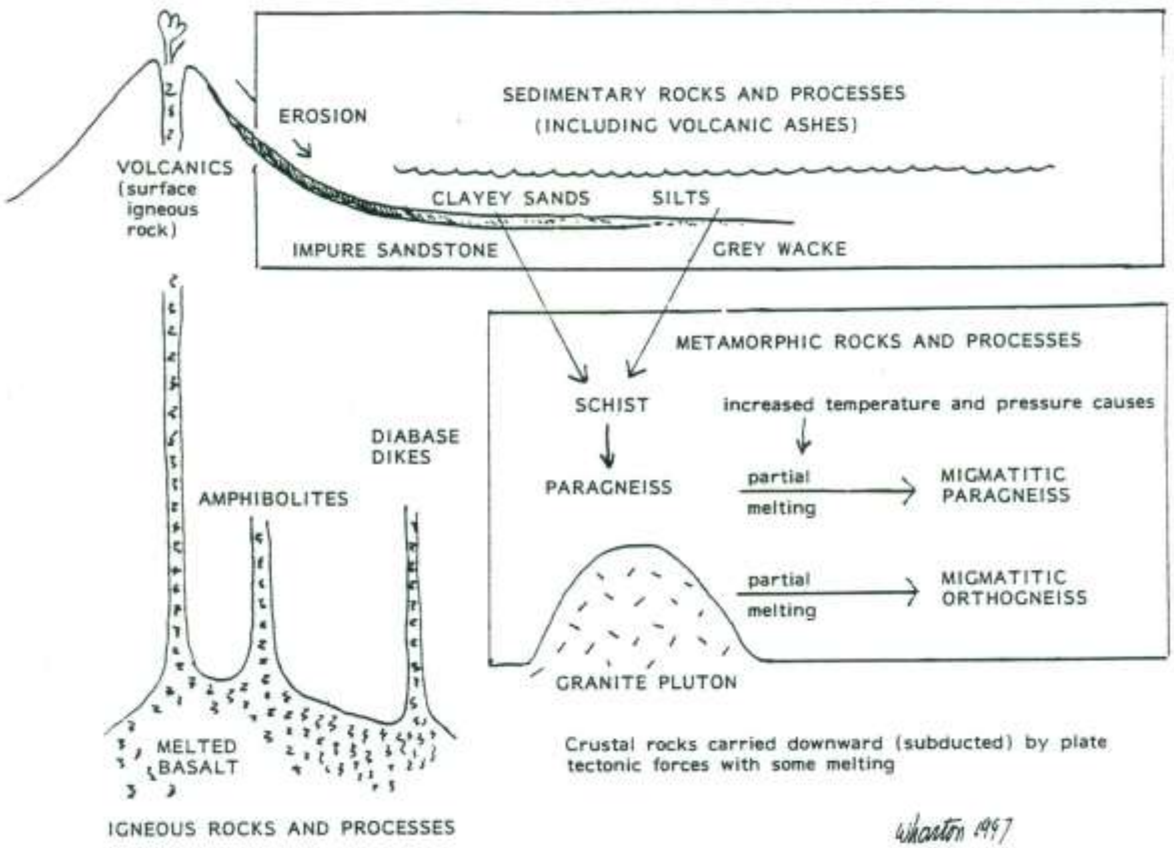


Fig. 2. Soil calcium levels in plant communities at SBG.

Fig. 2. Soil levels of calcium and plant communities at SBG. Ringed dots indicate circumneutral sites. Calcium in pounds per acre (divide by two to get ppm). Two important rock types are included for comparison.

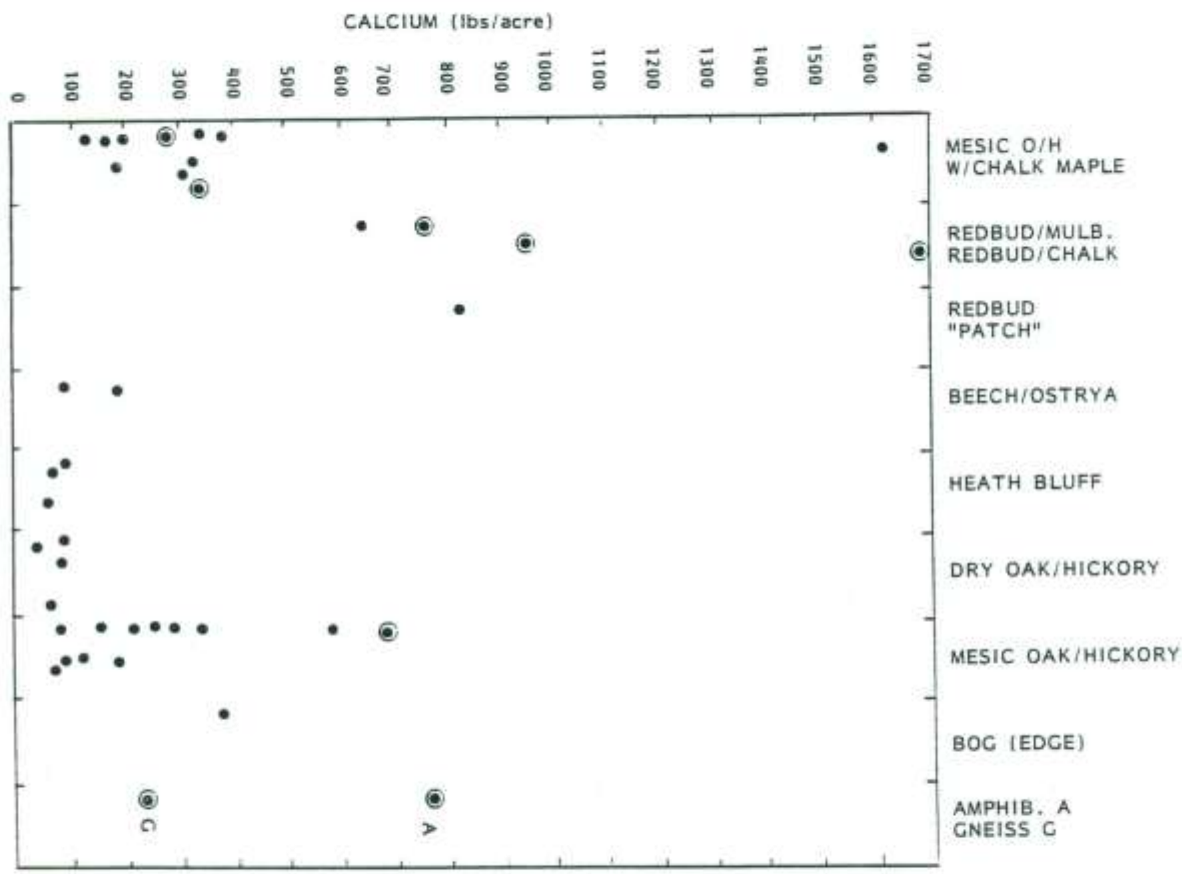
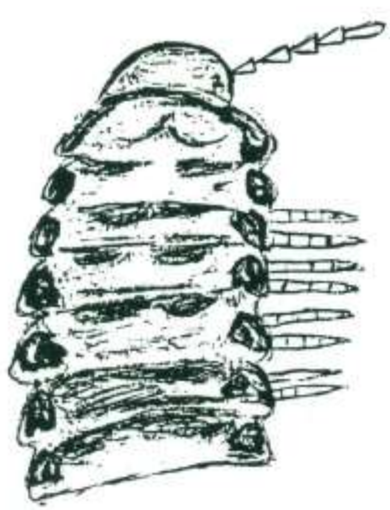


Fig. 3. Millipede color patterns.

Fig. 3. Approximate coloration pattern of two large Piedmont millipedes found at SBG. Animals are probably displaying warning coloration. Predators quickly learn their protective secretions of hydrocyanic acid.



Sigmoria rileyi

Cherokia georgiana

Fig. 4. Eight common millipedes of the SBG,

more or less drawn to natural size. Only a few of the body segments (tergites) are drawn for the "flat" Polydesmids. The first three species are round. They tend to curl up when disturbed. The five flat species have strong chemical defenses; some have warning colorations.

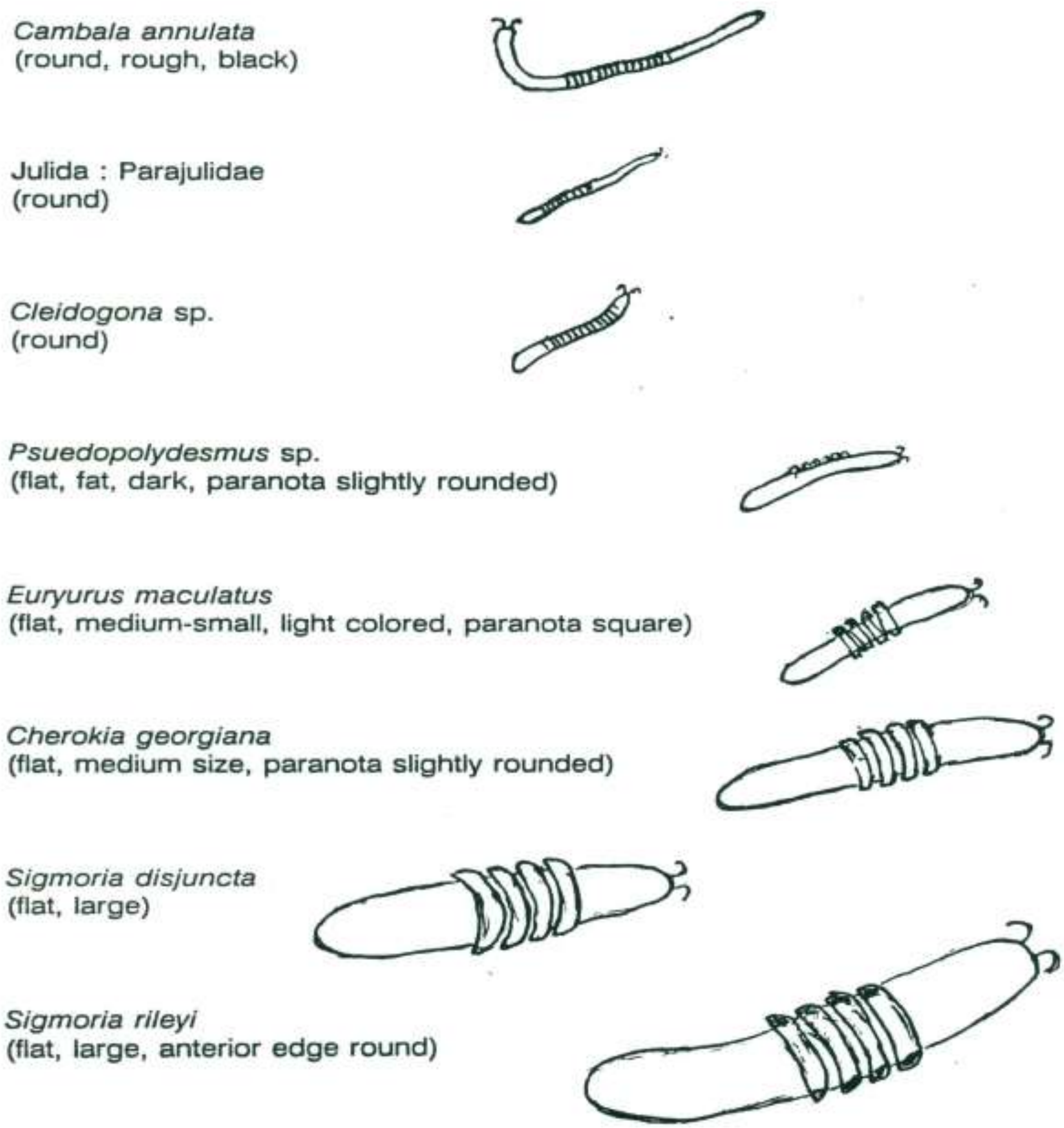


Fig. 5. Seven common larger snails of the SBG,

that apparently crawl through or on the leaf litter, or that graze on herbs or shrubs. With the exeption of the enlarged aperture of *Mesodon inflectus* to show the three "teeth," drawings are natural size. Dominant "Peterson characters" are indicated for quick identification in the field.

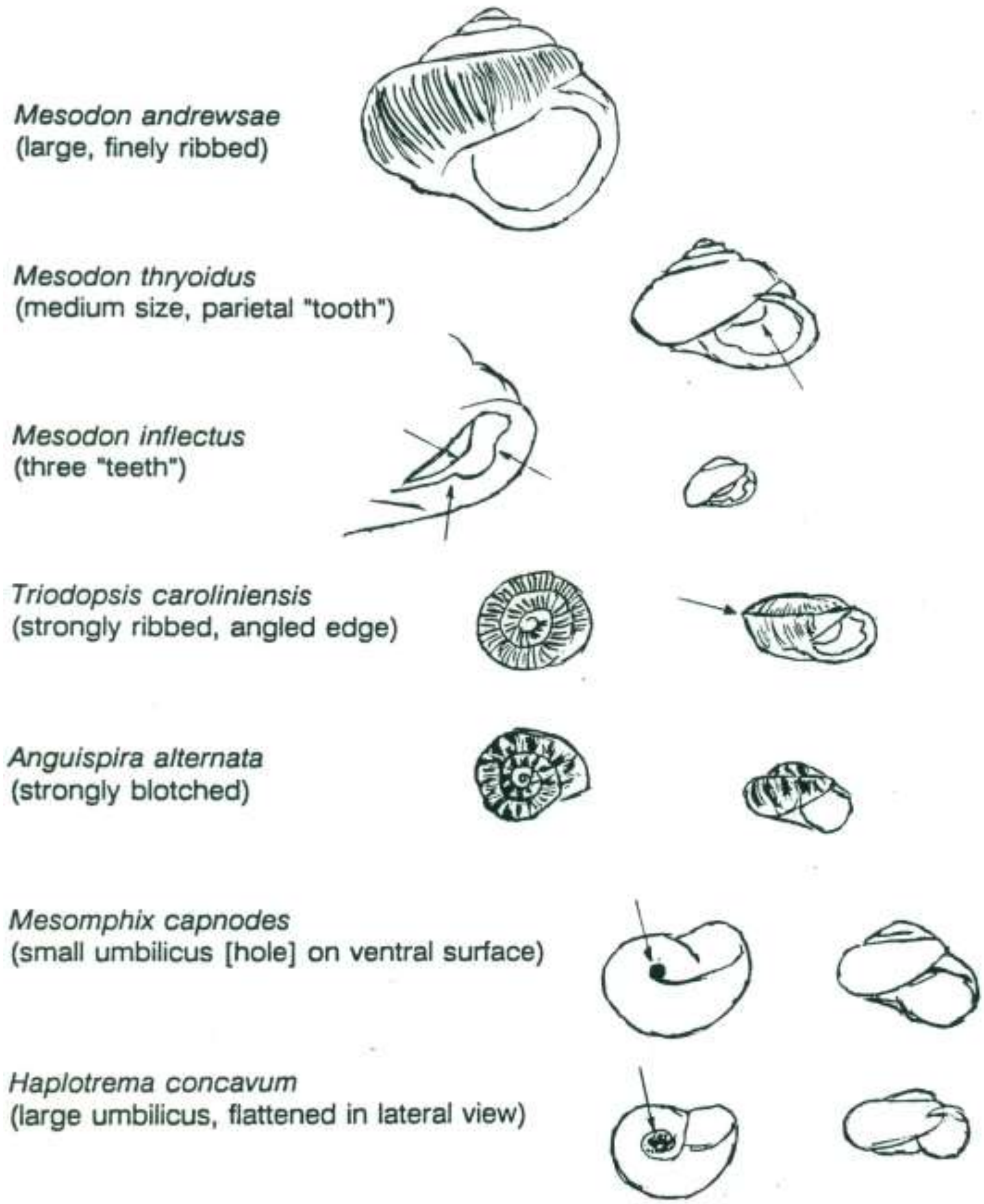


Fig. 6. Common summer resident birds of the SBG and the environments of plant communities where they are most likely to be seen or heard.

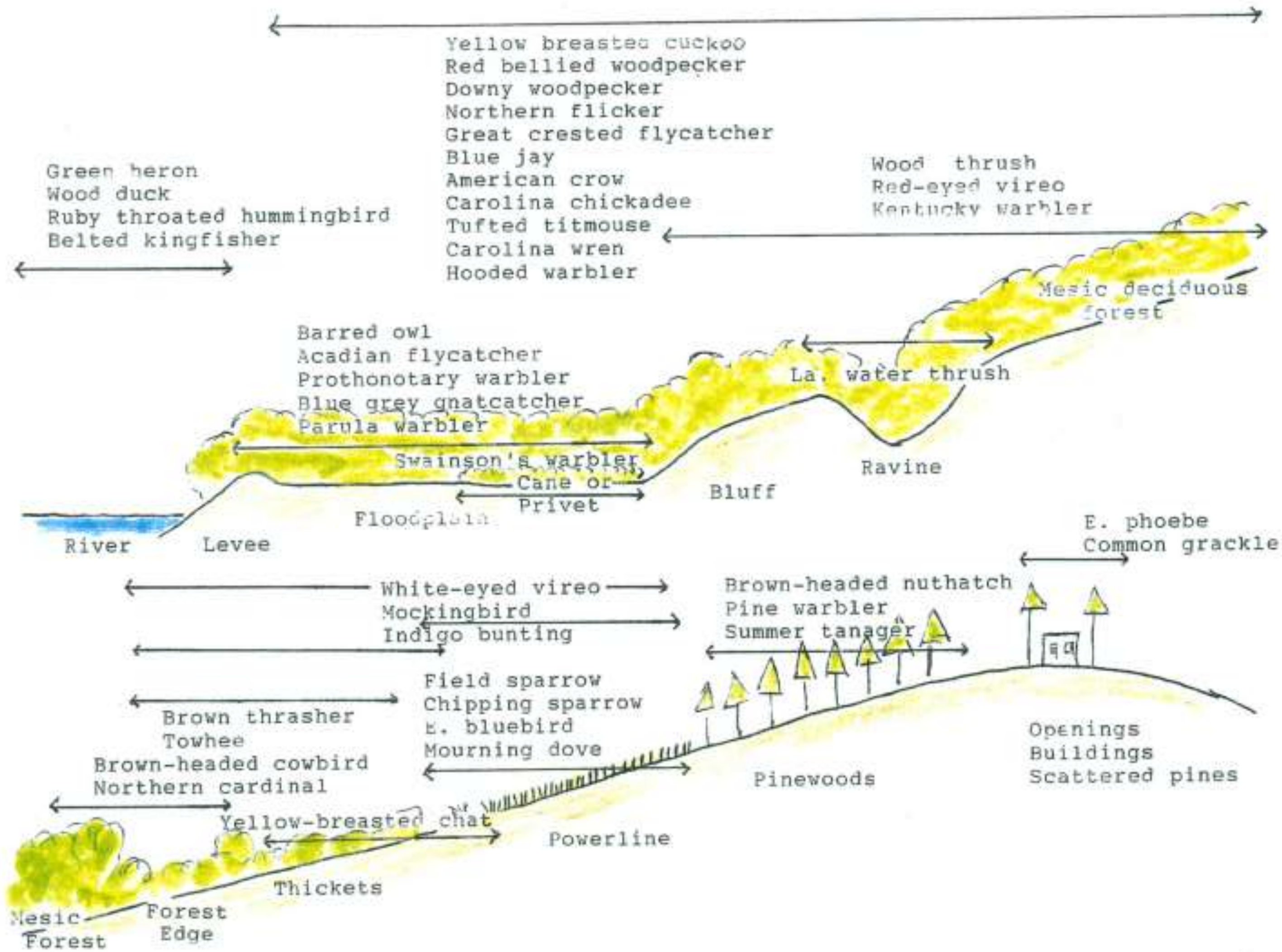


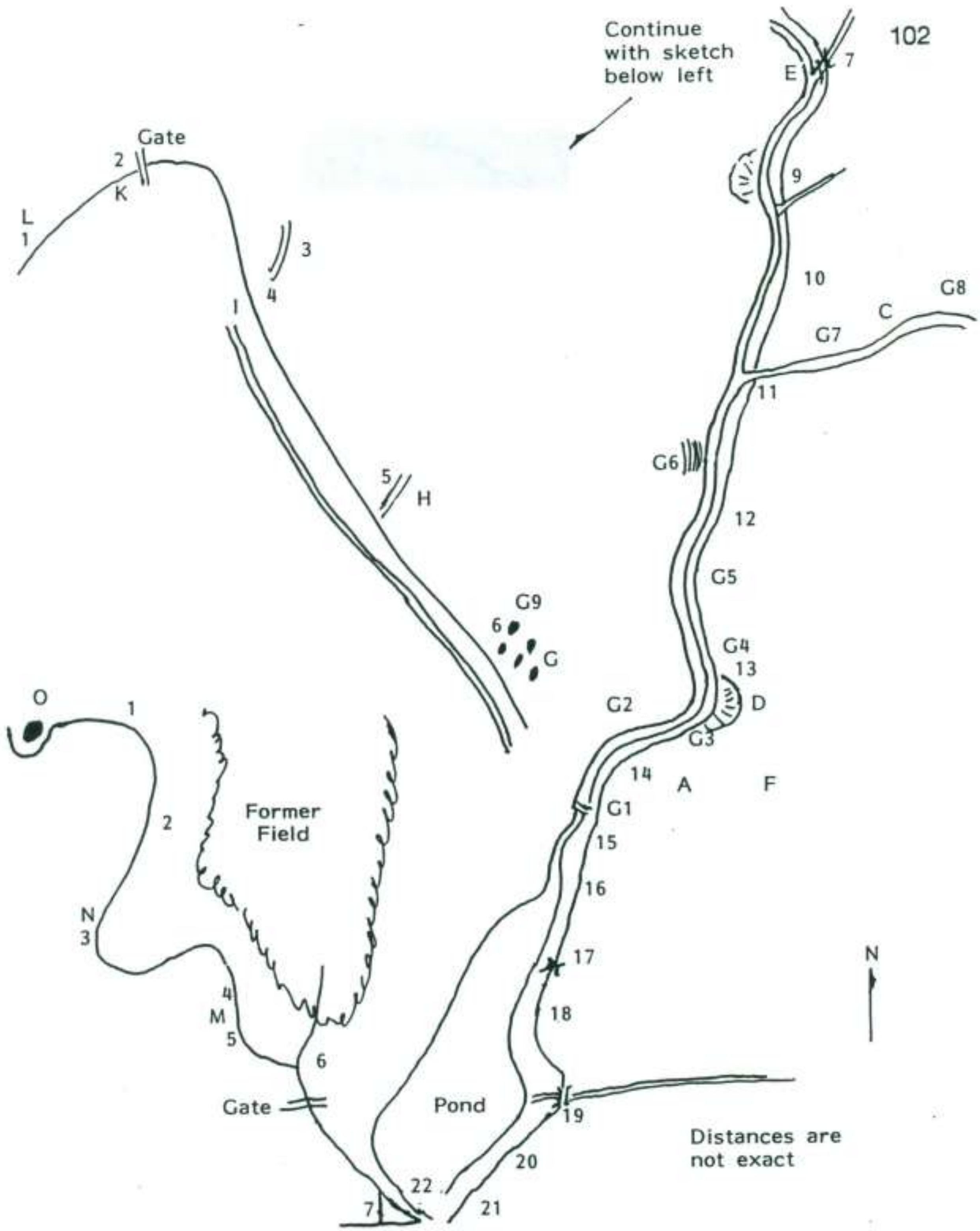
Fig. 6.Common summer resident birds of the SBG.

Maps

Map 1.Purple/Orange trail geologic stations and points of interest.

Map of geologic stations and points of interest along the Orange and Purple Trails. (Small numbers mark the approximate locations of wooden posts marked with numbers.) Geologic stations: G1, first waterfall (rock is migmatitic gneiss or "paragneiss"), below this point sand and silt deposits have covered the bedrock, creating a more or less flat floodplain, so that the streambed is no longer visible; G2, where creek turns east we see the first squarish boulders of amphibolite in creekbed. Just upstream the creek flows through an almost square "joint trough" in gneiss rock. G3, between these two station G3 markers is the first large outcrop of amphibolite crossing trail and dipping under creekbed. G4, here, and upstream, rocks in creekbed appear black due to a coating of manganese oxide, causing them to resemble amphibolite in color. G5, 70' south of marker 12 is a five foot block of foliated (layered) and weathered amphibolite in the creekbed (strike about N 15° E). G6, across the creek is an outcrop of fine-grained, homogenous (slightly migmatitic) gneiss; parallel foliations are clearly visible. G7, approximately 150' east of the fork (at the bench) a broad outcrop of amphibolite about 40 feet wide appears to cross Amphibolite Creek. In the bend at "C" it appears in the bank and float rock is common both in and out of the creek. Ga, 100' further east and on the north face of the ravine is a big outcrop showing intricate metamorphic folding in migmatitic gneiss. G9, northeast of the trail are numerous rockpiles (some are around tree bases), evidence of pre-cotton era subsistence agriculture.

Letters A through O are other points of geologic interest. A, boulders of migmatitic biotite gneiss. B, outcrop on hillside 75' above trail. Migmatitic gneiss showing excellent metamorphic folding (biotite mica forms dark bands). Look over the top and you will see some folds have weathered so that you can see daylight between them. These plunging folds strike S 30° W. C, amphibolite juts out in creek bank, represents east edge of 40' wide outcropping of amphibolite which begins 150' east of the junction of Amphibolite Creek with South Creek. O, a colluvial flat, probably formed by an old stream meander cutting into the hillside. E, large gneiss outcrop. F, modern human artifacts Ounkpile) on ridge. G, rockpiles (see text for explanation). H, erosion gullies from upslope agriculture. I, textbook example of headward erosion (how ravines and valleys are formed). Note also here a huge white oak on the far side of the ravine. Its root system is exposed and gives us the opportunity to contrast the aerobic zone of the soil (red from ferric iron) with the anaerobic (lacking in oxygen) zone (grey from ferrous iron) deeper below the roots (roots assist in reducing the iron). J, twin erosion gullies, possibly an old roadbed. K, enter zone of recent monoculture, probably cotton; L, edge of agriculture or "cotton terraces" can be seen to right of trail. M, amphibolite float rock has been found in trail between markers 15 and 16 (Orange Trail) and at N near marker 3 (Purple Trail). O, this large outcrop capping the hill is migmatitic paragneiss showing classic weathering by exfoliation giving the rounded appearance.



Continue
with sketch
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102
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G5
G4
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G2
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14
A
F
G1
15
16
17
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19
20
21
22
Pond
Distances are
not exact
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Gate
2
K
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Former
Field
N
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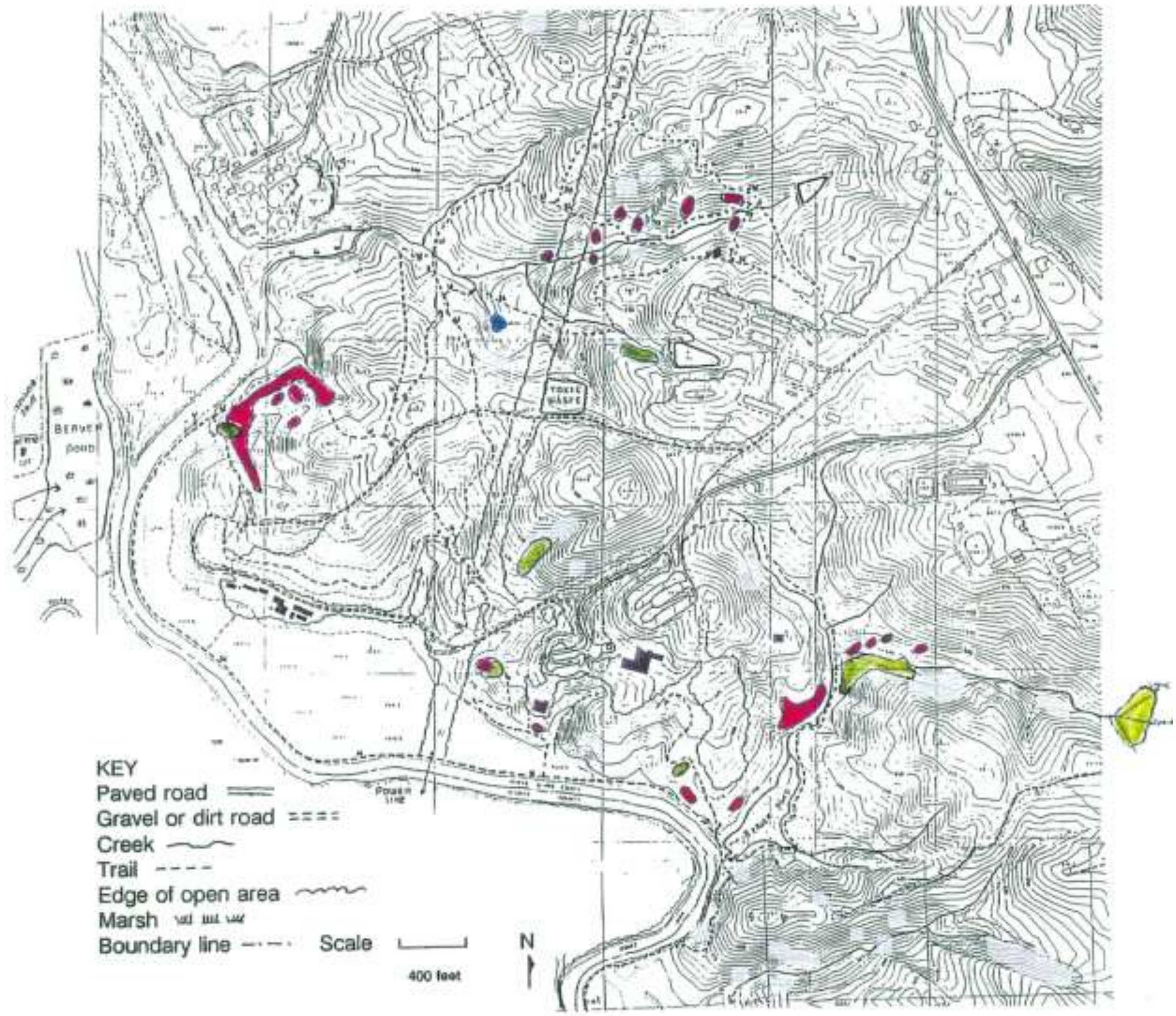
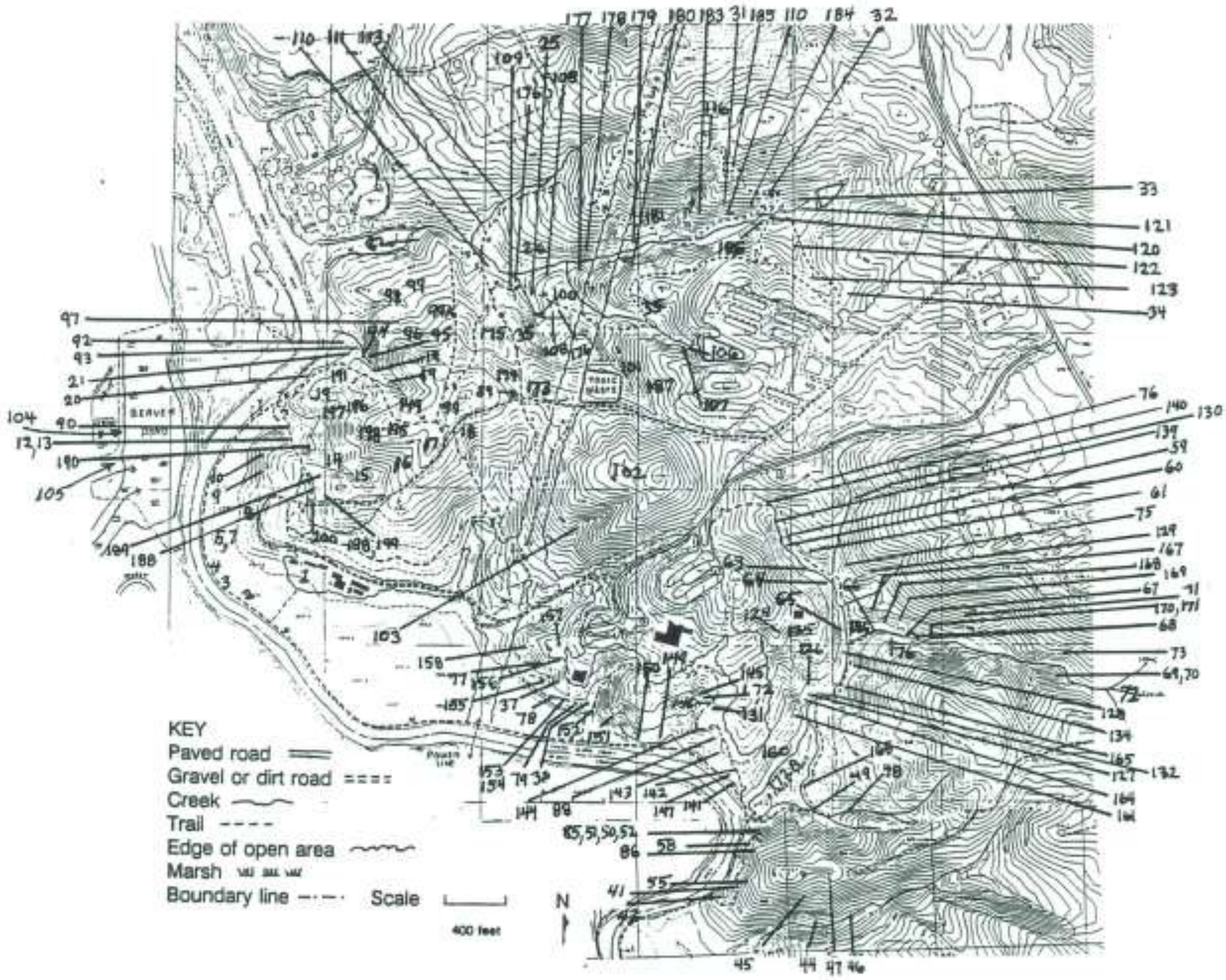
4
M
5
6
Gate
7

Map 2. Distribution of Chalk Maple and Beech; Bog

Distribution of chalk maple understory (pink) in mesic oak-hickory forest; beech (green) dominant or co-dominant in mesic oak-hickory forest; bog (blue).

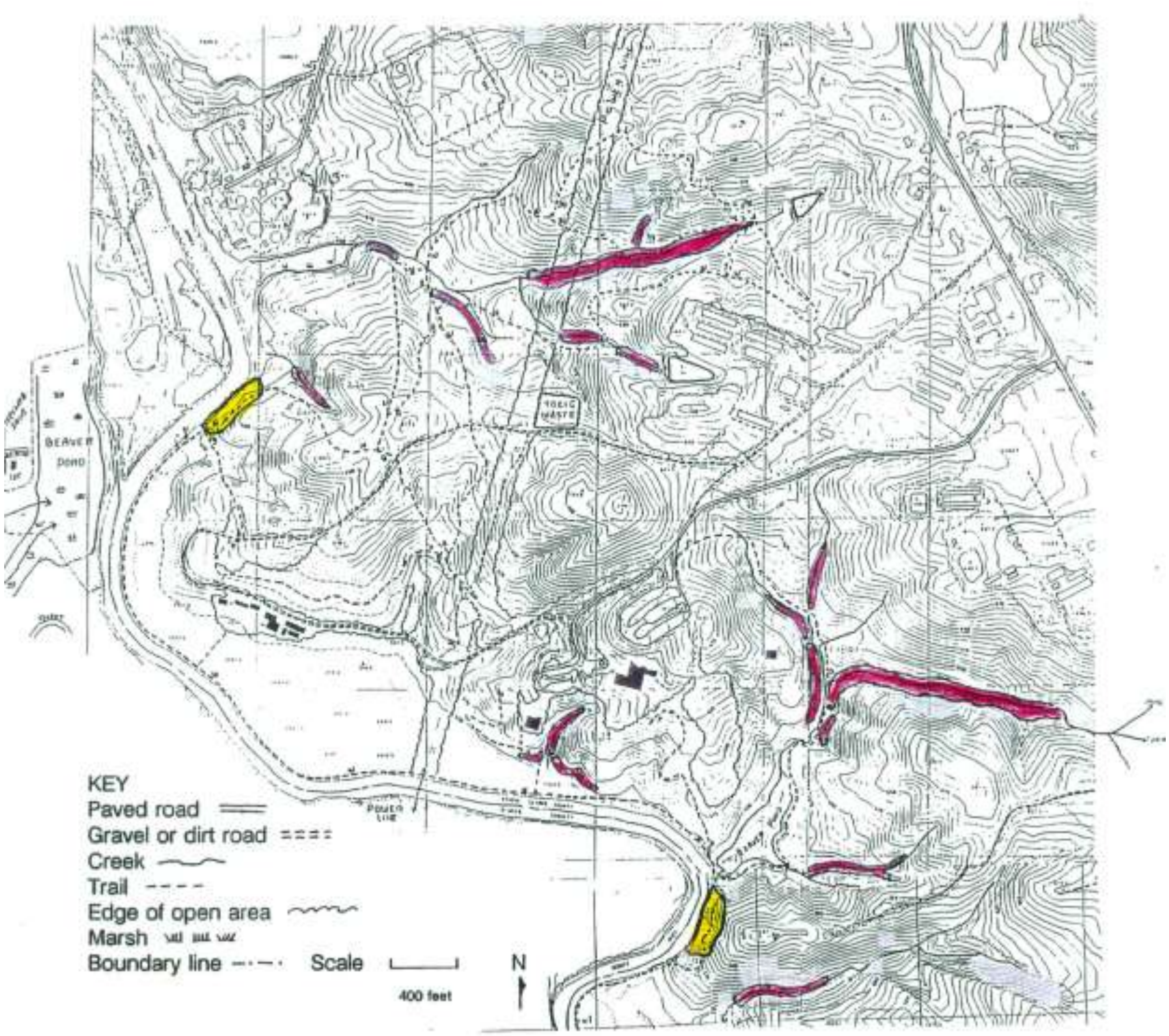


Map 3. Station numbers listed in Appendix 2.
Many of significance are discussed in Appendix 2.



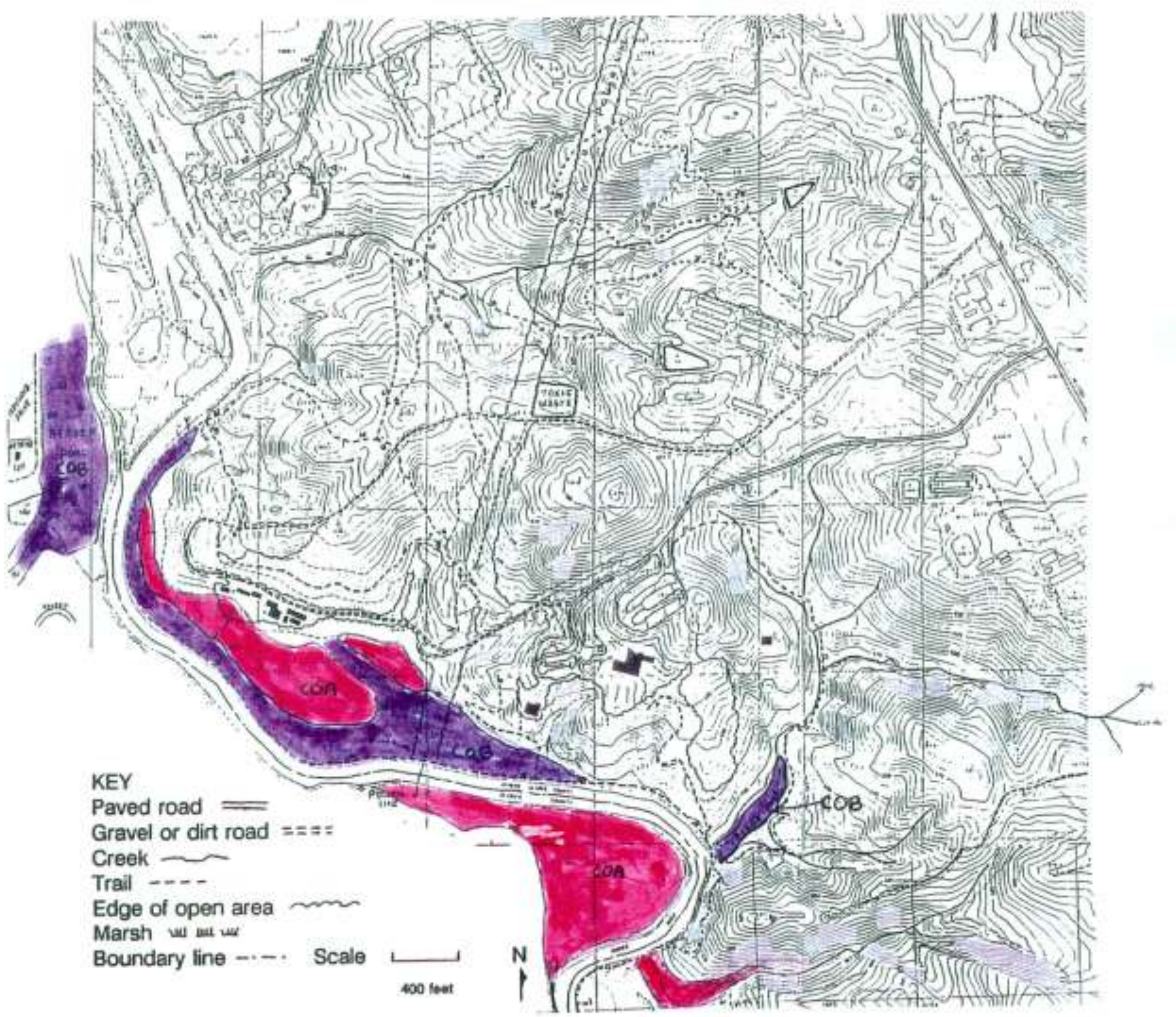
Map 4. Distribution of heath bluffs, ravines and lower (toe) slopes.

Heath bluffs (yellow); ravines and lower (toe) slopes (pink).



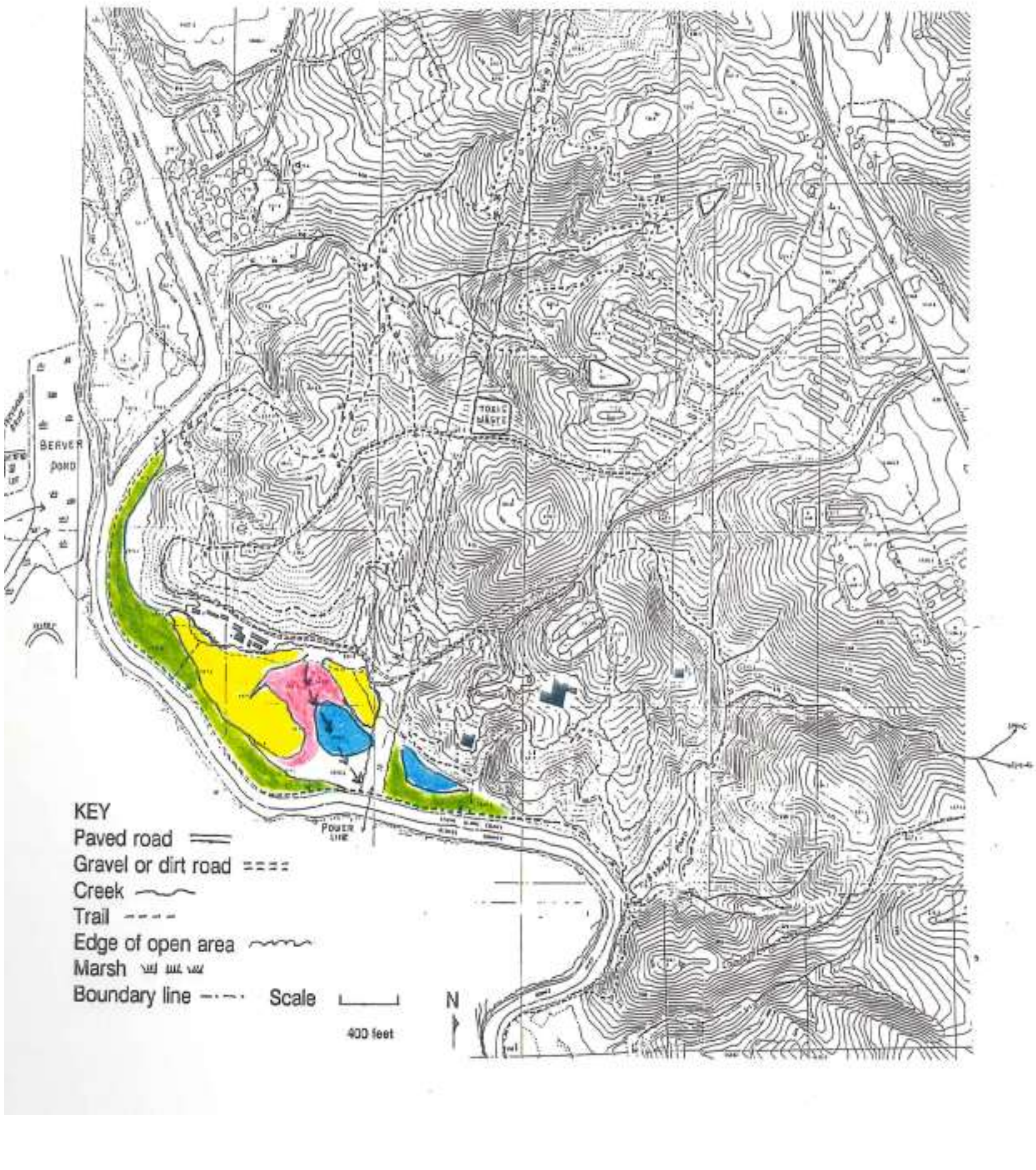
Map 5. Areas of floodplain soils.

Congaree (CoA), pink; Chewacla (CoB), purple. From NRCS Soil Survey, 1968.



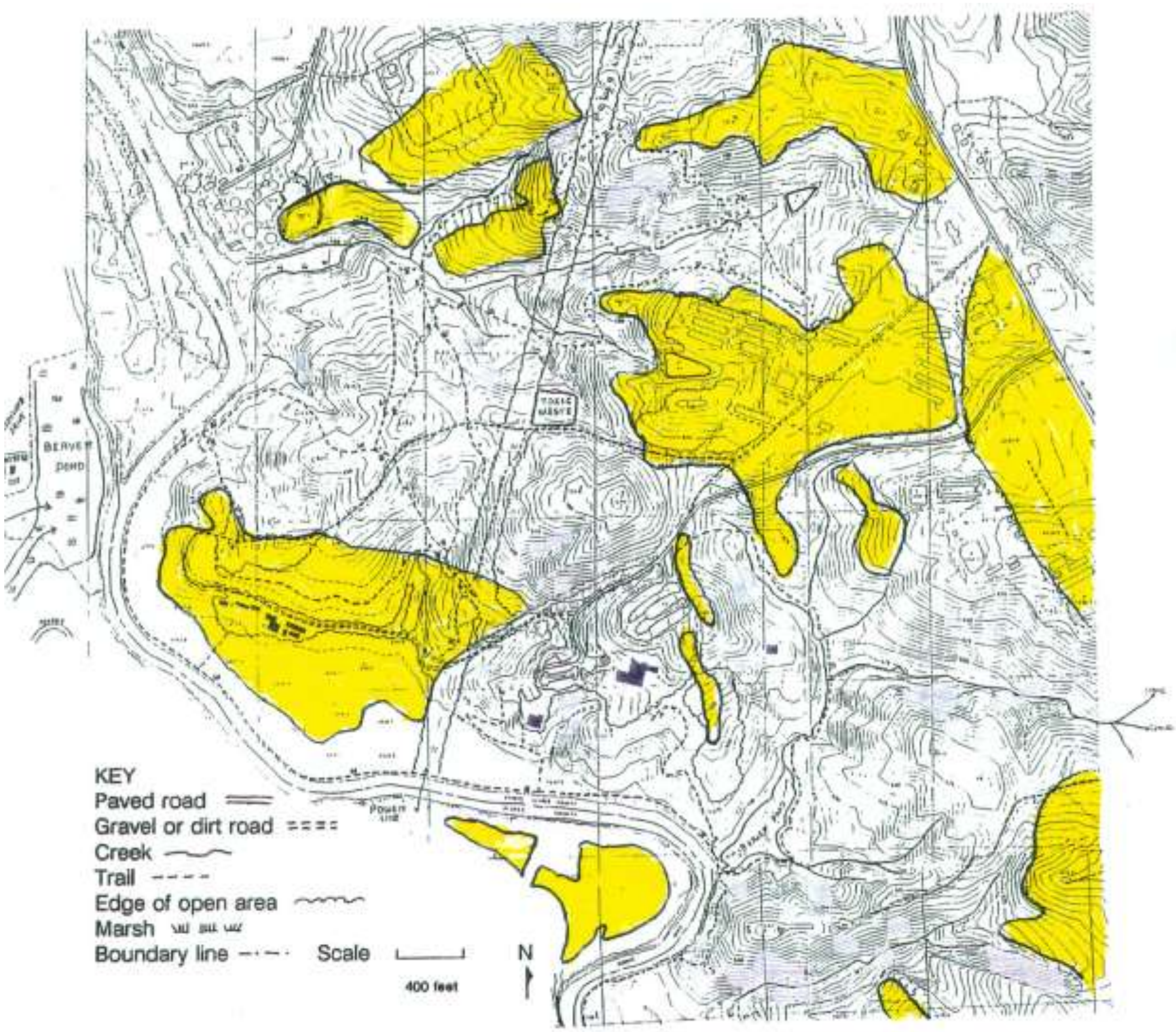
Map 6. Approximate vegetative patterns on the Oconee floodplain.

Some approximate vegetation patterns on the floodplain of the Oconee River (from aerial photographs and limited ground reconnaissance). Yellow: pine (comparatively recently in agriculture); Pink: very young bottomland hardwoods; Green: mature bottomland hardwoods, mostly with Japanese privet understory, often with standing water. Arrows indicate a principal drainage channel.



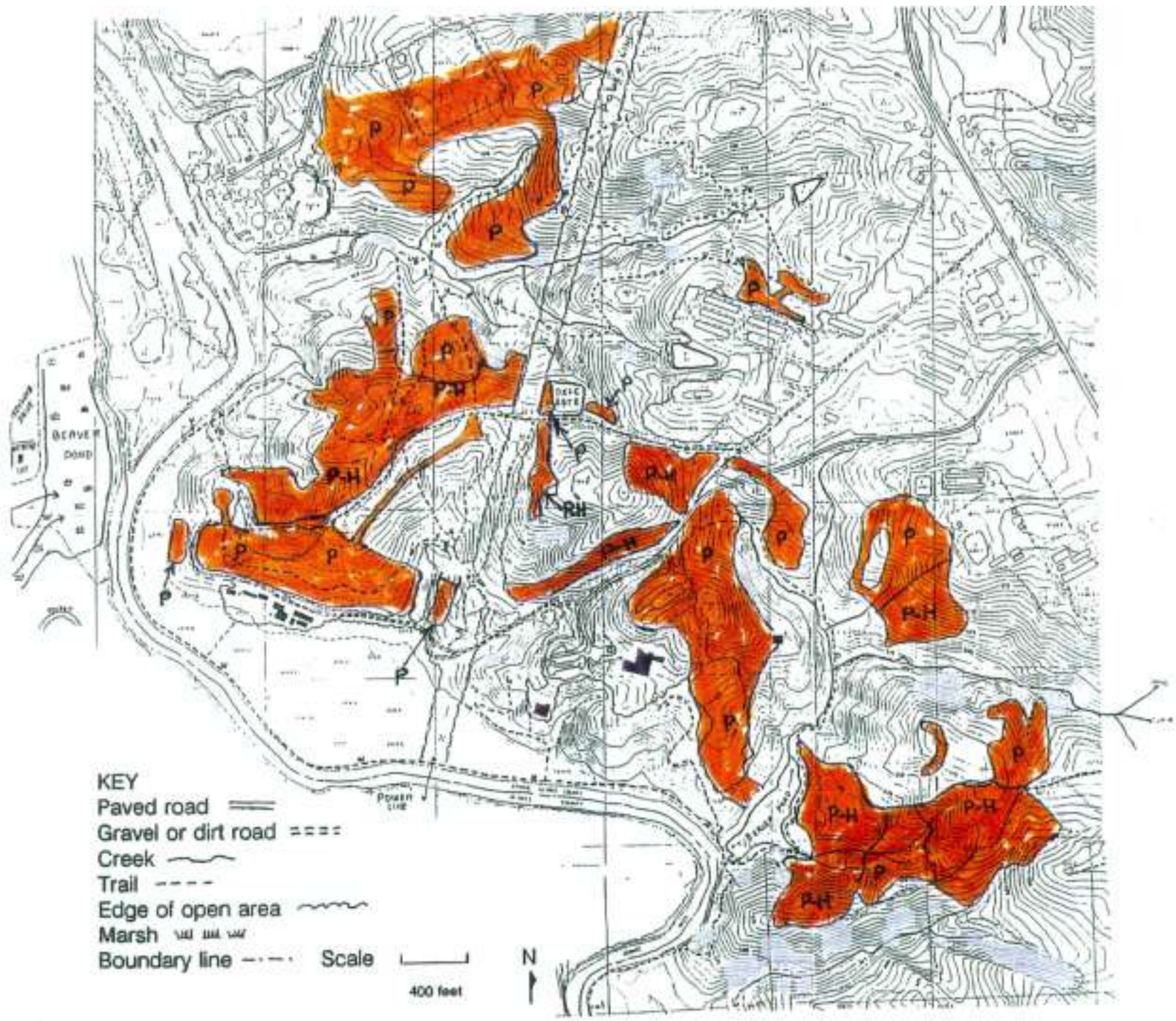
Map 7. Areas modified by man circa 1938.

Areas (yellow) in, or formerly in, agriculture which appear to have been in agriculture or otherwise modified by man circa 1938. From aerial photograph, courtesy Jim Affolter.



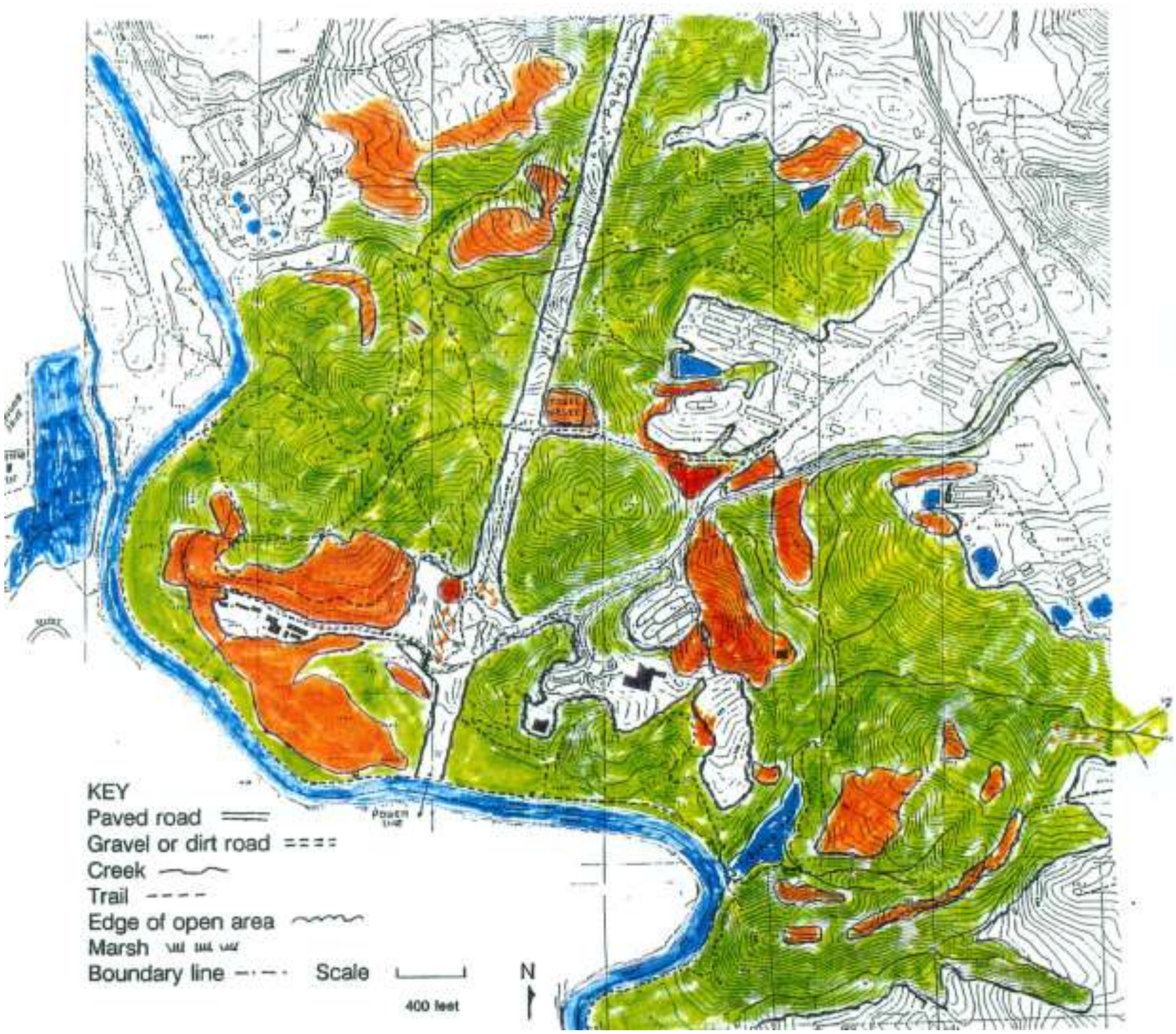
Map 8. Location of successional stages with pines in the upland.

Approximate location of successional stages with pines on the upland. Young to intermediae-aged pines (orange). Young to intermediate-aged (P). Mature pines (probably logged) and young hardwoods (P-H). Modified from William Beery (Vegetation and successional stages, UGBG map).



Map 9. Extent of canopy where pines are dominant or co-dominant (with hardwoods).

Extent of canopy where pines are dominant or co-dominant (orange) and extent of present open land (white). Water bodies are in blue. Based on a 1986 aerial photograph and recent infra-red photography, courtesy of Jim Affolter.



Map 10. Areas of critical importance impacted by human activities.

E, erosion; EPO, erosion and pollution from hog farm; P, excessive ponding on floodplain from upland runoff, timber kill anticipated; PO, pollution; PPO, potential pollution.